

CMSC 423: Computational Genomics

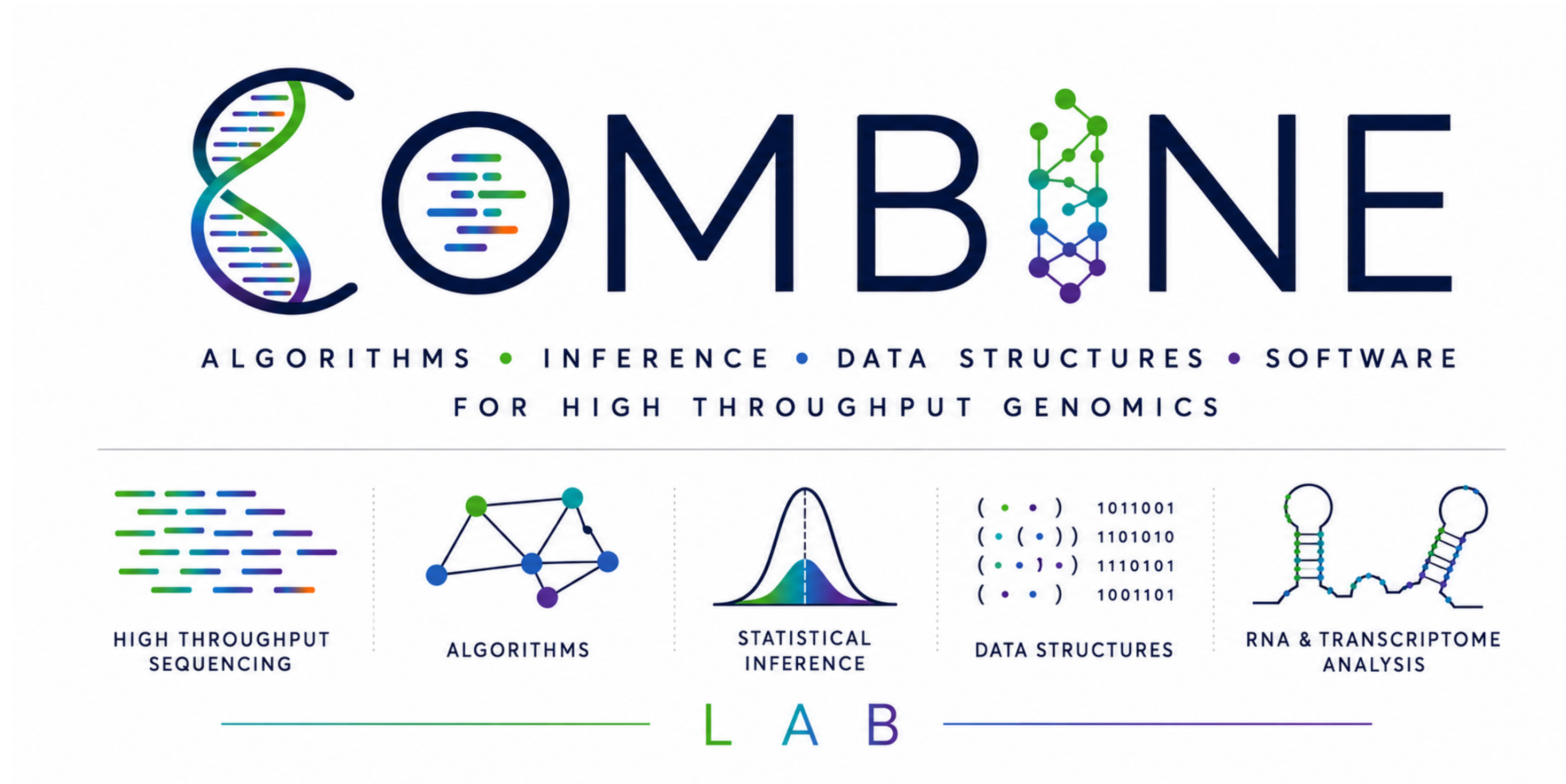
Fall 2026

Course Info

Instructor: Rob Patro (nomad@umd.edu)

Office: 3220 IRB

Course Website: <https://umd-cmsc423.github.io/f2026/>



Lab website: <https://combine-lab.github.io>

Course Info

TA:

Ming Gao

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Office hours: TBD (AVW 4140)

Course Info

ADS: <https://www.counseling.umd.edu/ads/>

Academic Integrity: <https://academiccatalog.umd.edu/undergraduate/registration-academic-requirements-regulations/academic-integrity-student-conduct-codes/>

Piazza Page: <https://piazza.com/umd/fall2026/cmssc423>

Gradescope: <https://www.gradescope.com/courses/1354177>

If you have a class-related e-mail: Please **prefix the subject with [CMSC423-F26]**, so that my filter will pick it up and it won't be accidentally routed to SPAM.

Coursework & Grading

Coursework and grading: The coursework will consist of a number of different programming projects, two midterm exams and a final exam. The breakdown of weights for these different assignments will be as follows:

- Programming assignments - 30%
- Midterm 1 - 20%
- Midterm 2 - 20%
- Final Exam — 30%

Late policy: Assignments that are turned in late will be docked 1% for each hour they are late up to the first 48 hours. The policy for “rounding” hours is in the student’s favor (i.e., the penalty will be truncated to the nearest complete hour). **After 48 hours, late assignments will not be accepted.**

Regrade policy: All requests to re-grade, re-check, or re-mark an assignment or exam question **must be made in writing**. When the assignment is re-graded, it will be re-checked in its entirety. This means that *it is possible to lose points on other problems if they were graded incorrectly or too leniently the first time*. Therefore, I urge you to thoroughly consider each regrade request you make.

Academic Integrity

maintain it!

TLDR : Don't cheat. Don't copy code from friends, classmates, or the internet for the short programming assignments or the projects. Don't provide code to classmates for any of the assignments or projects. Don't cheat on the exams. Be cool, and everything will be cool.

Academic integrity is a very serious issue. Any assignment, project or exam you complete in this course is expected to be your own work. If you are allowed to discuss the details of or work together on an assignment, this will be made explicit. Otherwise, you are expected to complete the work yourself. *Plagiarism is not just the outright copying of content.* If you paraphrase someone else's thoughts, words, or ideas and you don't cite your source, this constitutes plagiarism. It is always much better to turn in an incorrect or incomplete assignment representing your own efforts than to attempt to pass off the work of another as your own. **If you are academically dishonest in this course, you will receive a grade of XF, and you will be reported to the university's Office of Student Conduct.**

Textbooks

Based on student feedback from previous offerings of this course, there *is no required text*.

The course website lists several *optional* textbooks (some free) that you may find useful, stratified by topic (algorithms, biology, etc.)

Additional material will be made available on the course website as needed.

However, this is an *upper-level* course, and you should *absolutely* seek out other sources explaining these topics from different angles, using different notations and examples, etc.

If you seek out other sources and still are having difficulty with an idea, *please* reach out to us (myself & Ming). Also, consider reaching out to your peers *via* Piazza.

Other Resources

Software development and the command line

While you should have had, in the prerequisites for this course, exposure to the relevant tools (e.g. how to properly create a tarball, how to invoke the compiler from the command line / a script and how to use tools like Git), I realize it may have been a while since you have used these skills. If you feel you need a refresher on these topics, I would strongly suggest checking out ["The missing semester"](#) (an MIT course dedicated to these various miscellaneous topics).

The “workhorse” bioinformatics algorithms are implemented in tools that are *command line applications*. They follow (often) traditional Unix design principles, are invoked from the command line, read files as input and produce other files as output.

Hence, our projects will be designed to work in a similar way.

Basics you should refresh

- File and directory structure and navigation
- File movement, copying, compression (tar + gzip), etc.
- Working in a **compiled** language (C, C++, Go, Rust, Java, etc.)
- Building software **from the command line** (bash script, Makefile, CMake, Cargo, etc.)
- (If choosing Java): Making a Java program act as a “normal” executable
- Reading command line parameters (passing parameters and options to your program)
- Reading and writing of files (reading files from disk, writing output to disk)
- Serialization and Deserialization of data structures (saving data structures to disk & reading them back)

What to expect from this course

A few notes about CMSC 423 this semester:

Some material is inherently advanced—but it is also powerful and broadly useful. Ask questions; I will do my best to make sure everyone can follow.

Some projects are intentionally challenging. You will have time to work through them, and we will give substantial feedback.

Implementing—and understanding—the projects is one of the best ways to prepare for the exams. The exams ask you to reason about the algorithms and data structures we study.

From genomes to algorithms

How do we turn billions of noisy, fragmented observations into biological knowledge?

BIOLOGICAL QUESTION

What are we trying to learn?



SEQUENCING DATA

What does the experiment actually observe?



COMPUTATIONAL METHOD

Which model, algorithm, and data structure connect evidence to an answer?

By the end: recognize this structure—and choose or design an appropriate method.

A rose by any other name

bioinformatics vs. computational biology

Search filters: All, Images, News, Videos, Shopping, More, Tools

About 35,400,000 results (0.91 seconds)

<https://www.medicaltechnologyschools.com> › bioinfor...

Bioinformatics vs. Computational Biology: A Comparison

As both fields rely on the availability and accuracy of datasets, they usually help one another reach their respective project goals. While **computational** ...

People also ask

- What is the difference between bioinformatics and computational biology?
- Is bioinformatics better than computational biology?
- Is computational biology a good field?
- What can I do with a masters in computational biology?

Feedback

<https://www.northeastern.edu> › graduate › blog › comp...

Computational Biology vs. Bioinformatics: What's the Difference?

May 28, 2021 — While Kaluziak notes that there is a great deal of overlap between **computational biology** and **bioinformatics**, the latter requires programming and ...

<https://www.reddit.com> › bioinformatics › comments

Computational Biology versus Bioinformatics - Reddit

Jan 22, 2016 — There's considerable overlap, but in general **Computational Bio** is more concerned with developing the theory and writing the software to answer ...

Should we draw a semantic distinction between the terms bioinformatics and computational biology? Are the differences substantial enough and the definitions well enough agreed upon ?

Yes; use them differently

51.6%

No; use interchangeably

48.4%

1,415 votes · Final results

7:55 PM · Jun 29, 2022 · Twitter for Android

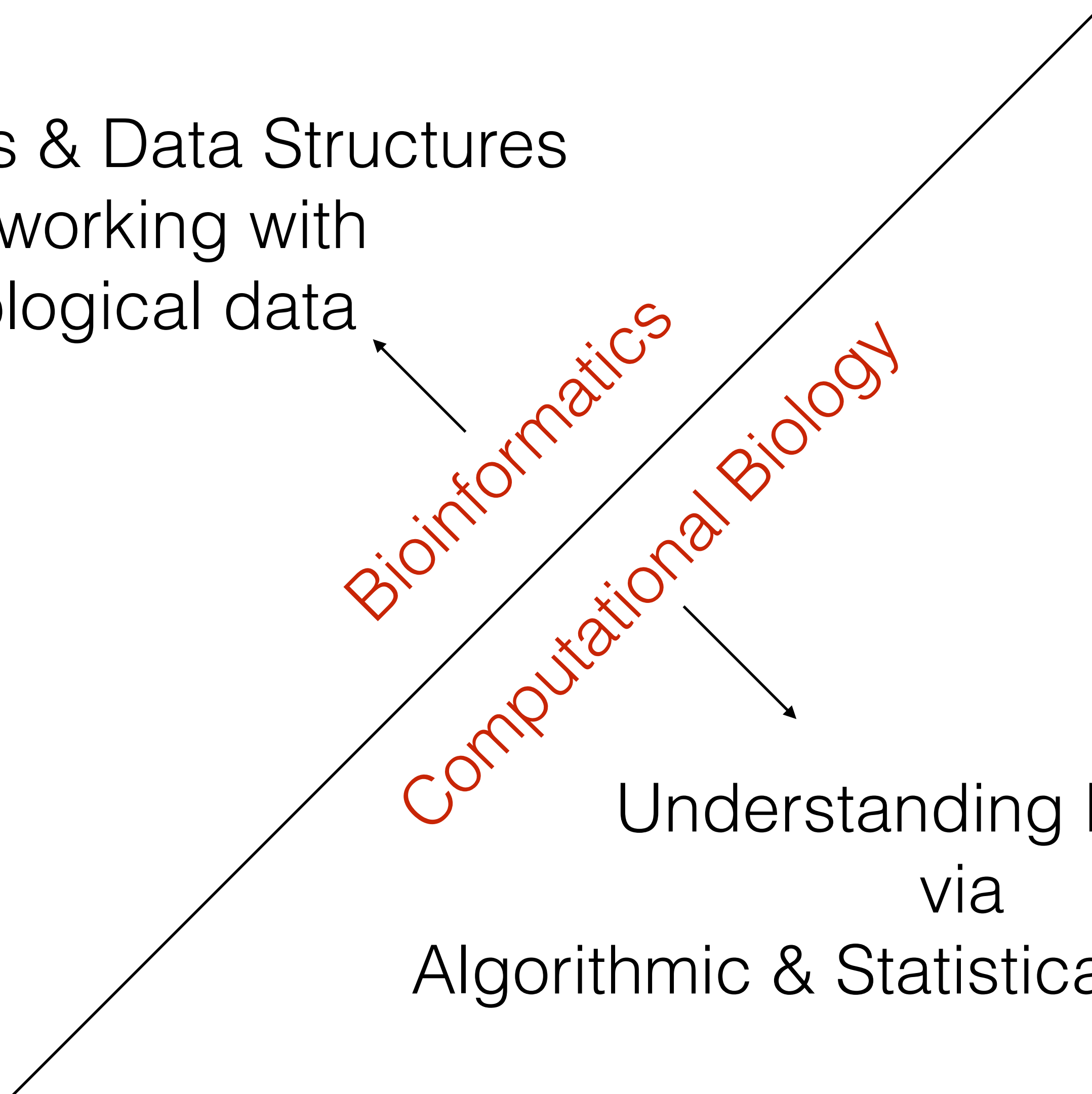
Bioinformatics & Computational Biology

Algorithms & Data Structures
for working with
Biological data

Bioinformatics

Computational Biology

Understanding Biology
via
Algorithmic & Statistical Approaches



Bioinformatics & Computational Biology

We'll treat this as two sides of the same coin
&
try to ignore this distinction

Why Bioinformatics (genomics)?

Our ability to measure biological data at high throughput has been transformative

1990 - 2000

Sequencing the first human genome took ~10 years and cost ~\$2.7 **billion**

Today

Sequencing a genome costs ~\$100 - 1,000* (depending on how you count)

~18 Tb per “run” at maximum capacity

Progression of sequencing capacity

\$3,000,000,000	2003 Human Genome Project	
\$20,000,000	2006 1 st individual genome	
\$2,000,000	2007 1 st NGS Genome	
\$200,000	2008 1 st 30x genome	
\$10,000	2010 1 st sub-10K genome	
\$1,000	2014 1 st \$1,000 genome	
\$100	2017 1 st \$100 genome	

Tons of Data, but we need Knowledge

We'll discuss a bit about how sequencing works soon. But the hallmark *limitations* are:

- Short “reads” (75 — 250) characters when the texts we’re interested in are 1,000s to 1,000,000,000s of characters long.
- Imperfect “reads” — results in infrequent but considerable “errors”; modifying, inserting or deleting one or more characters in the “read”
- Biased “reads” — as a result of the underlying chemistry & physics, sampling is not perfectly uniform and random. Biases are not always known.

A cartoon view of sequencing

Orig. genome: GACGATGAAGCCAGCTCGGCTGAACACGATTCCAACGTCTAACGCTCGGTAGGTCTGGCAGGTAGCCGGGCGGAA

Many copies: GACGATGAAGCCAGCTCGGCTGAACACGATTCCAACGTCTAACGCTCGGTAGGTCTGGCAGGTAGCCGGGCGGAA
GACGATGAAGCCAGCTCGGCTGAACACGATTCCAACGTCTAACGCTCGGTAGGTCTGGCAGGTAGCCGGGCGGAA
GACGATGAAGCCAGCTCGGCTGAACACGATTCCAACGTCTAACGCTCGGTAGGTCTGGCAGGTAGCCGGGCGGAA
...
GACGATGAAGCCAGCTCGGCTGAACACGATTCCAACGTCTAACGCTCGGTAGGTCTGGCAGGTAGCCGGGCGGAA

“random”
fragmentation
into short
sequences:

ATGAAGCCAGC
TCTAACGCTCG
GATTCCAACGT
TAACGCTCGGTAGCTCTGGC

TTCCAACGTCTAA
AACGCTCGGTAGGTC
CGATGAACCCAGCT

CCAACGTCTAAC
CGTCTAACGC
ATTCCAACGTCTAAC

GTCTAACGCTCGGTAGGTC
TGAAGCCAGCTCG
TAAGTCTGGCAGG
TCTGGCAGGGAG

ACGATGAAGCCAGCTCG
CGTCTCACGC

CTGATCACGATTCCAACGT
CTGAACACGATTCCA
TTCCAACGTCTAGCGCTTGG
GCTCGGCTGAACACG

GACCATGAAGCCAGCTCGG
TCTGGCAGGT

AAGCCAGCTCGGCTGAACA
GCTCGGCTGAACACTATT
TAGGTCTGGCA

TAACGCTCGGTAGGTCTGGC
TGTACACGATTCCGA
CGGCTGAACACGATTC
ACGATTCCAACGTCTAACG
CTCGGAAGGTCTGGCAGG

sequencing “read”

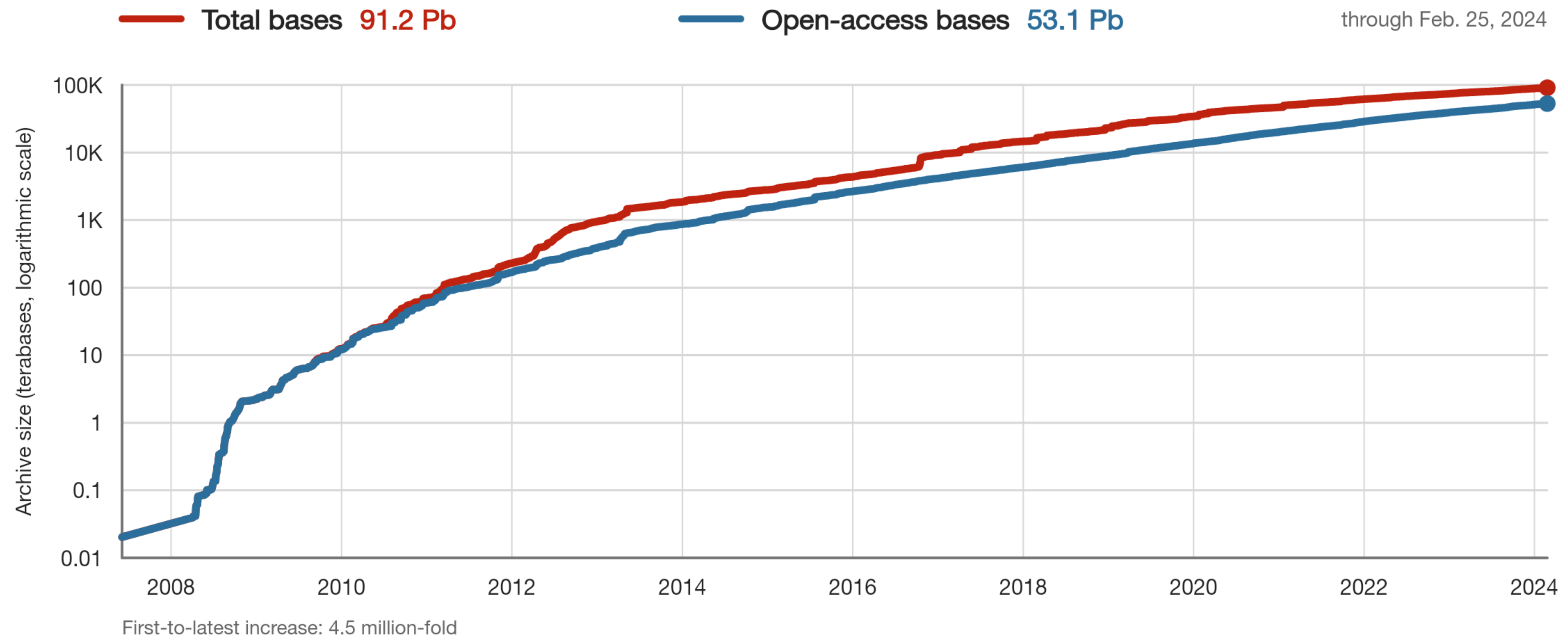
The FASTA format (relevant for Project 0)

record {
 header {
 identifier
 comment
 }
 sequence
}

```
>NM_001168316 comment_for_the_record
TTAGTGAGGTTGGGGAGAGATAACGCTGTAACTTTTATTTTTCAGGAAATCTGGAAACC
TACAGTCTCCAAGCCTGCTCAGCCAAGAAGGAGCTCACTGTGGGCACCAGAGACAGGGAC
CCAATGTGGAGACCTGTGAGCCTGTGTCCGGCCCTGAACTCTCAAGCACAGGGCAGGCTT
CCTGAGCATTGAAGAGAATATGTGGGAGAACAAAACAGAACTGAAAGAATATGCAAGGT
GTCTTTCTTGGATGTTATTCCATGATAGATAGTAGGGGCAGGAGTGAGAGAGGCTGACTA
GGTCTGGACATGGAGGCTGGAAGAGTCAGGGTGTGATTCGGAGAGGCGCATGAGAAGGAA
GGTGGATTTTAAGGCTGGAAATCTGAGGGTCAGTGGTCCAAGTCACTCAGAGACAGAATC
ACAGCATAGCCCTTGCTGATGGCAAACAAAGGAGGACAAGAGGACTGGAAAGAATTCTGC
TAGCAGGCAGGAGCTAGTAAGGATGAATTTGTAGCAAAATTAGCAAGTGGAAAGGATGAT
TTTTGGCCATTTTTCCTGTTCTTCAAGAAAACAGG
>NM_174914
TTAGTGAGGTTGGGGAGAGATAACGCTGTAACTTTTATTTTTCAGGAAATCTGGAAACC
TACAGTCTCCAAGCCTGCTCAGCCAAGAAGGAGCTCACTGTGGGCACCAGAGACAGGGAC
CCAATGTGGAGACCTGTGAGCCTGTGTCCGGCCCTGAACTCTCAAGCACAGGGCAGGCTT
CCTGAGCATTGAAGAGAATATGTGGGAGAACAAAACAGAACTGAAAGAATATGCAAGGT
GTCTTTCTTGGATGTTATTCCATGATAGATAGTAGGGGCAGGAGTGAGAGAGGCTGACTA
GG
>NR_031764 comments=optional
TTAGTGAGGTTGGGGAGAGATAACGCTGTAACTTTTATTTTTCAGGAAATCTGGA
>NM_004503 wrapping_width=variable
TTTTGTCTGTCCTGGATTGGAGCCGTCCCTATAACCATCTAGTTCCGAGTACAACTGGAGACAGAAATAAATATTAAAGAAATCATAGAC
CGTCGCCCTCAATTCCACCGCCTATGATCCAGTGAGGCATTTCTCGACCTATGGAGCGGCCGTTGCCCAGAACCGGATCTACTCGACTCCC
CAG
```


Despite these limitations, sequencing datasets have grown at a breakneck pace

The SRA grew ~4.5 million-fold in 17 years



Source: NCBI SRA growth statistics (through Feb. 25, 2024)

Like all technologies, biotech marches forward

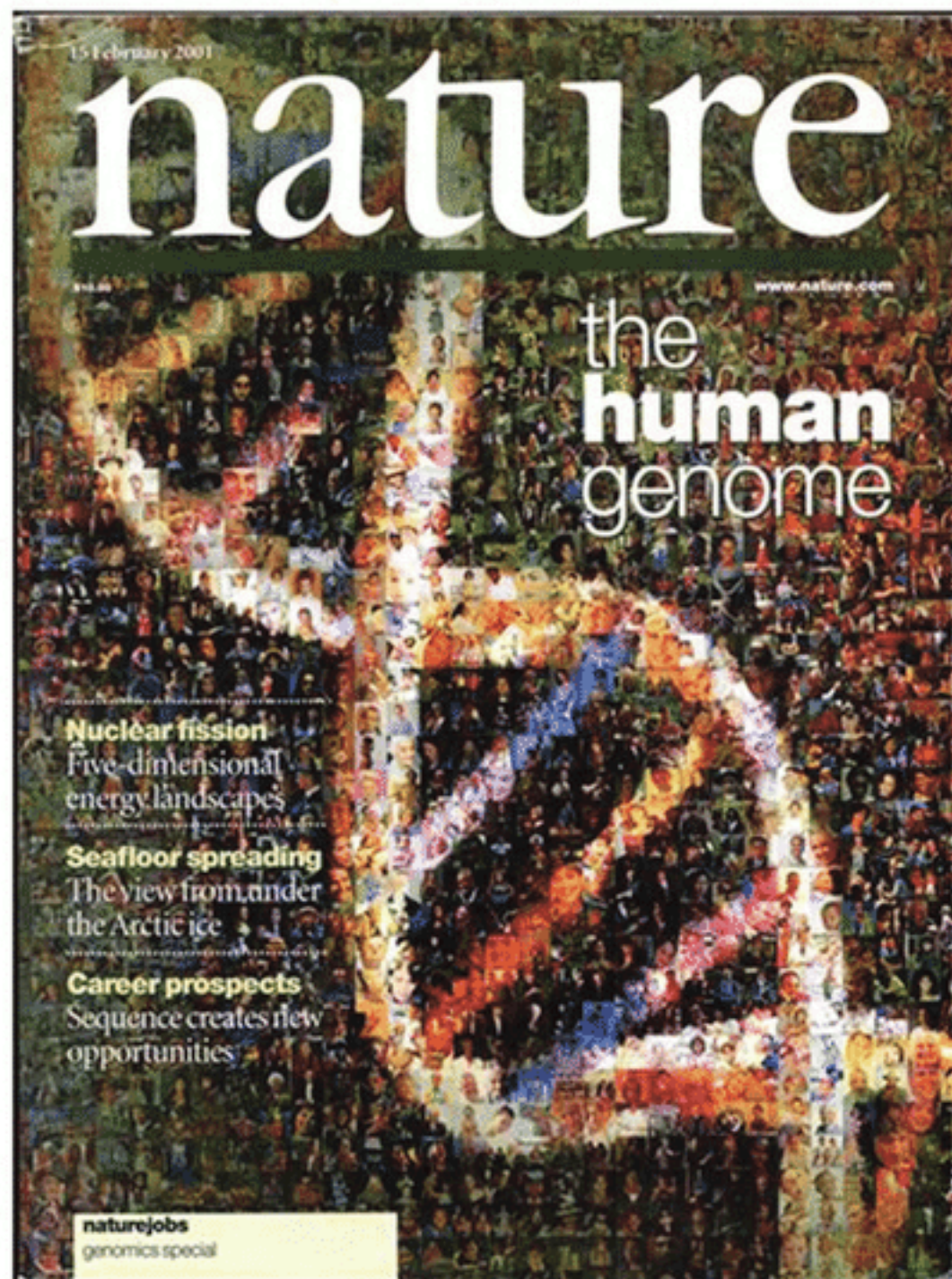
A lot of our focus will be on short-read technologies, but recent improvements in long-read technologies have been transformative:



- Much longer reads (10-25kb for PacBio HiFi, up to ~1MB for ONT)
- Many fewer independent samples (long reads, but fewer of them)
- ONT error rate higher than “short reads” 1-5%, though manufacturers claim $\sim < 1\%$, PacBio can match Illumina error rates but is substantially more expensive

Actually completing the human genome

2001



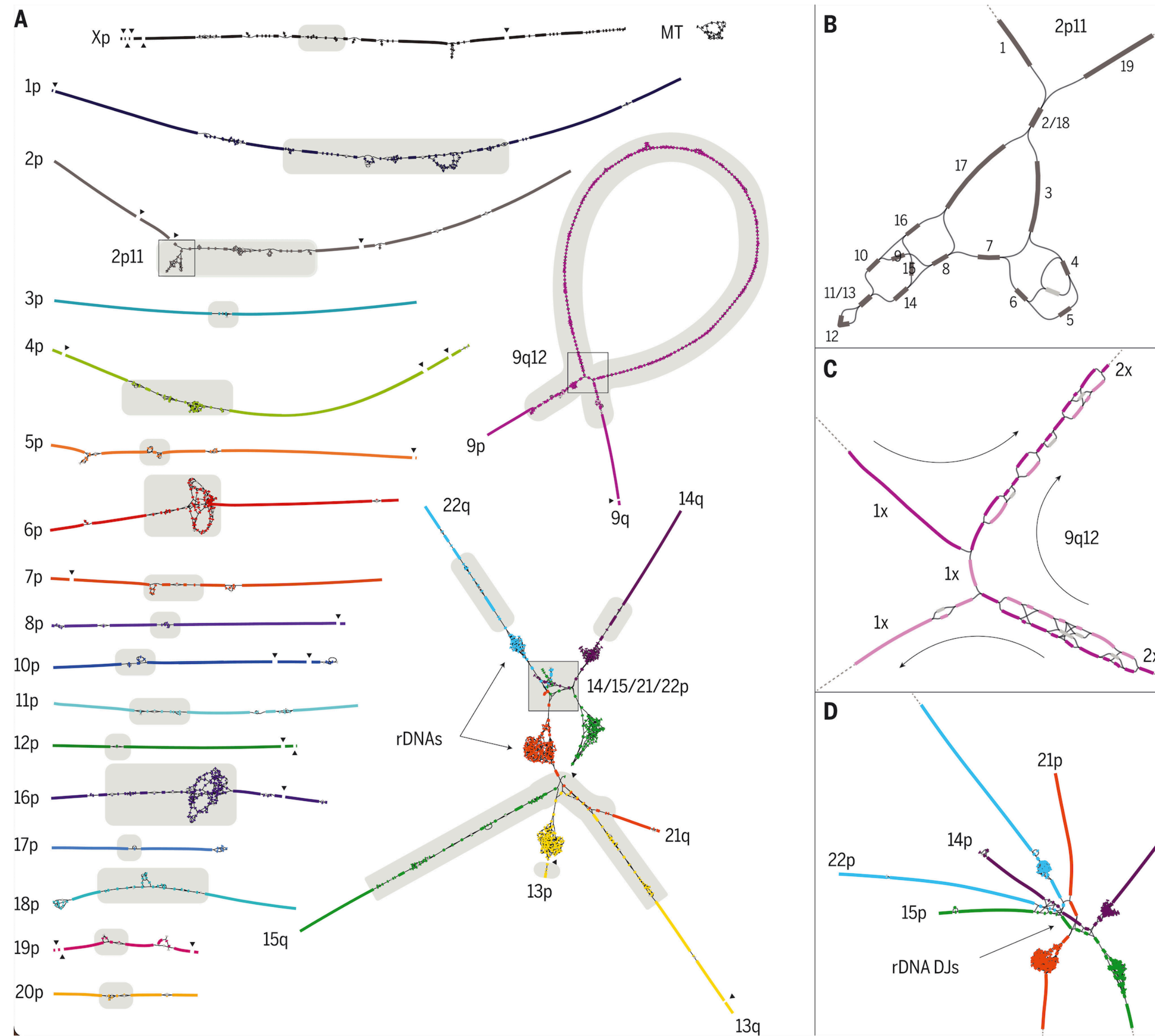
algorithmic
&
biotech
progress



2022



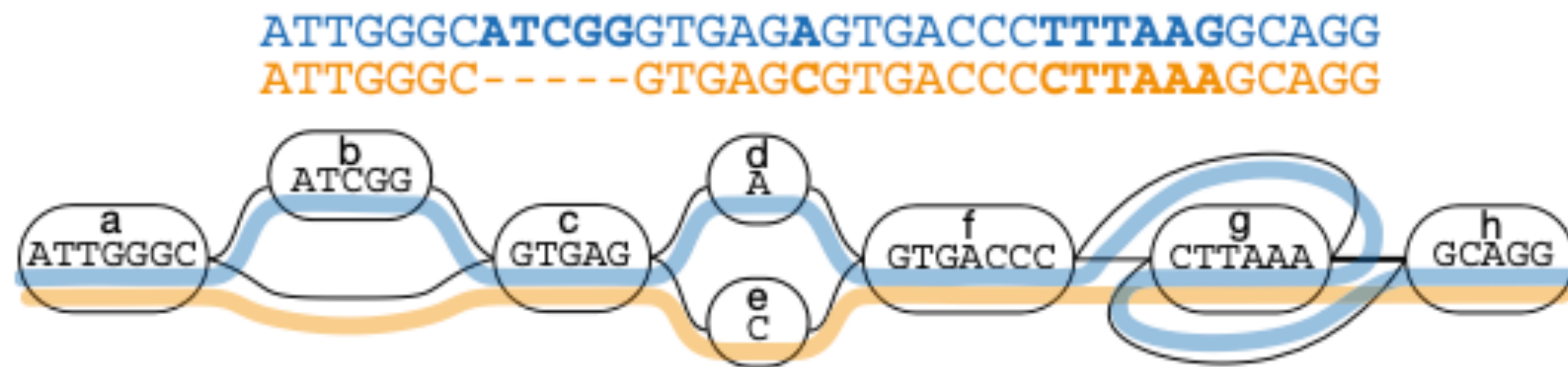
Assembly of the human genome



A complete genome is not a complete reference

Telomere-to-telomere assembly solved a long-standing challenge: a complete sequence from one human genome.

But one genome cannot represent human variation. The next step is a pangenome reference built from many diverse genomes.



Article

A draft human pangenome reference

<https://doi.org/10.1038/s41586-023-05896-x>

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Open access

 Check for updates

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Answer questions “in the large”

What is the genome of the terrapin? (genomics)

Which genes are expressed in healthy vs. diseased tissue? (transcriptomics)

How do environment changes affect the microbial ecosystem of the Chesapeake bay? (metagenomics)

How do genome changes lead to changes & diversity in a population? (population genetics/genomics)

How related are two species if we look at their whole genomes? (phylogenetics / phylogenomics)

Some Computational Challenges

Answering questions on such a scale becomes a *fundamentally* computational endeavor:

Assembly — Find a likely “super string” that parsimoniously explains 200M short sub-strings (string processing, graph theory)

Alignment — Find an *approximate* match for 50M short string in a 5GB corpus of text (string processing, data structure & algorithm design)

Expression / Abundance Estimation — Find the most probable mixture of genes / microbes that explain the results of a sequencing experiment (statistics & ML)

Phylogenomics — Given a set of related gene sequences, and an assumed model of sequence evolution, determine how these sequences are related to each other (statistics & ML)

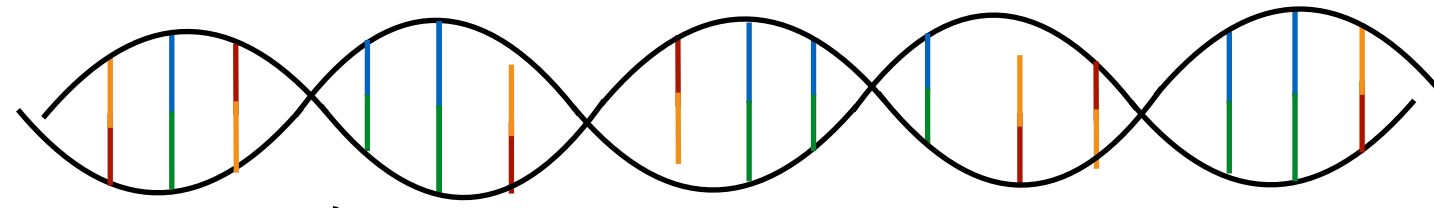
Establishing a basic lexicon

This course will focus on algorithms and data structures (with a little bit of probability & statistics), but the ***problems*** we will explore derive directly from biological questions.

In order to motivate our Computer Science, we will need a basic understanding of the Molecular Biology in which the problems are phrased.

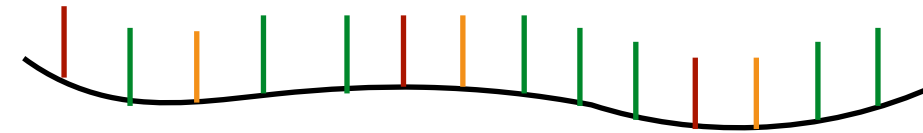
“Flow” of information in the cell

DNA



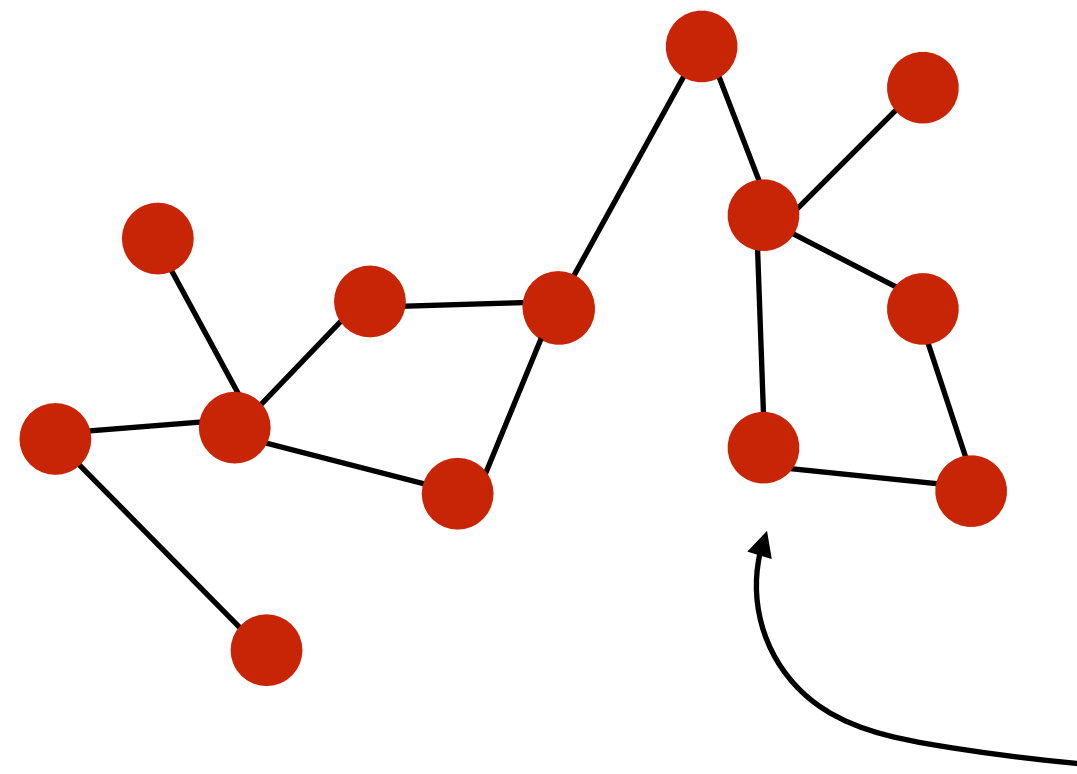
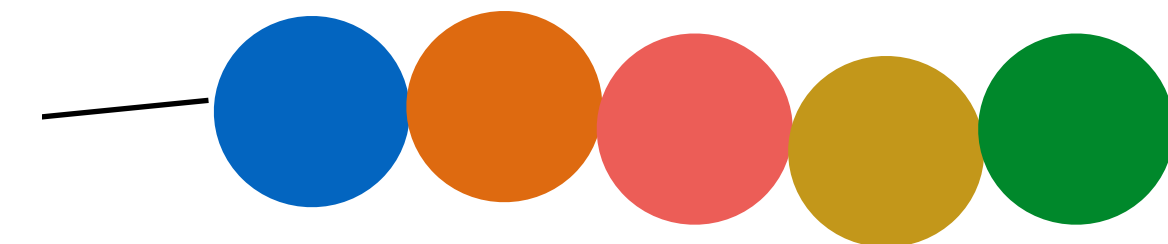
RNA Polymerase
(transcription)

RNA



Ribosomes
(translation)

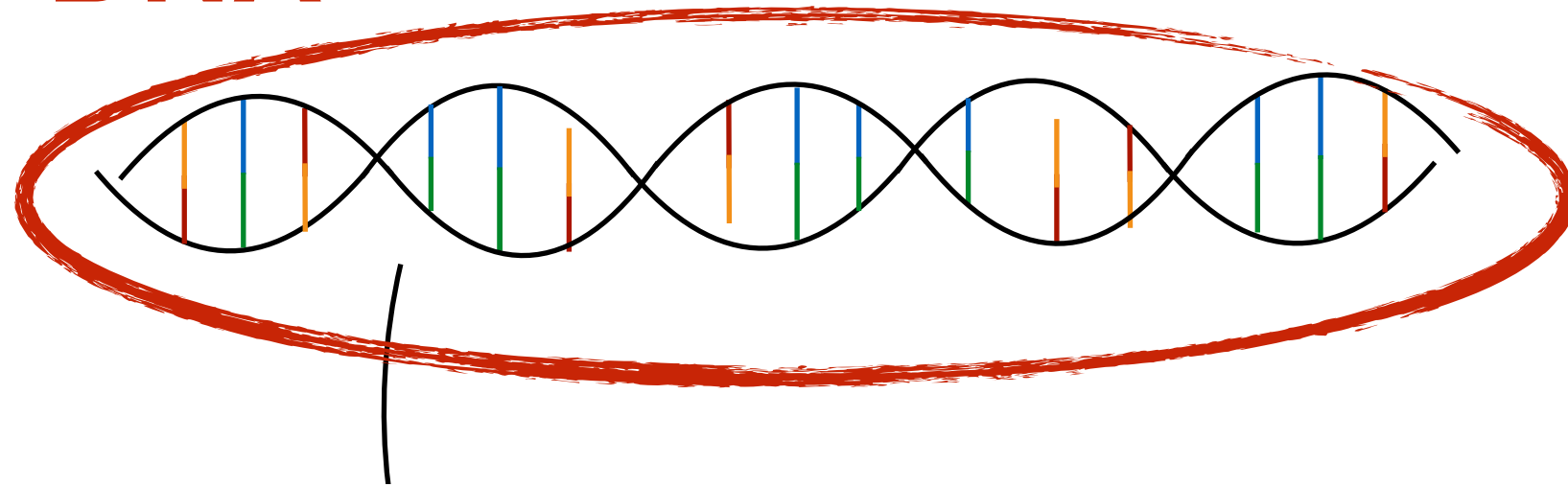
Protein



Form networks &
pathways; perform a
vast set of cellular
functions

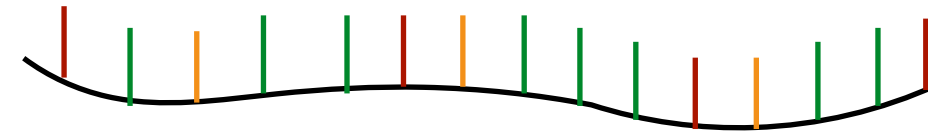
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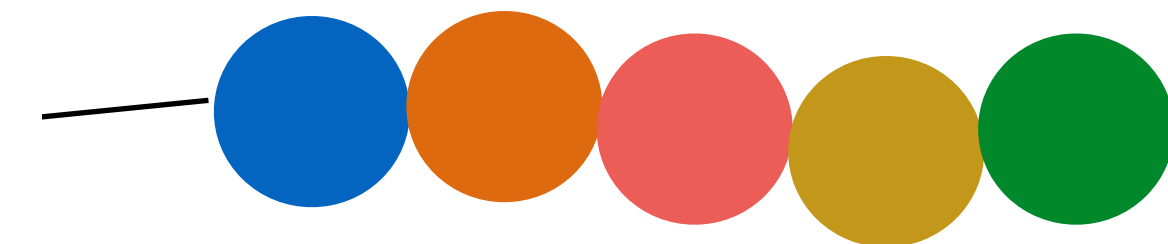
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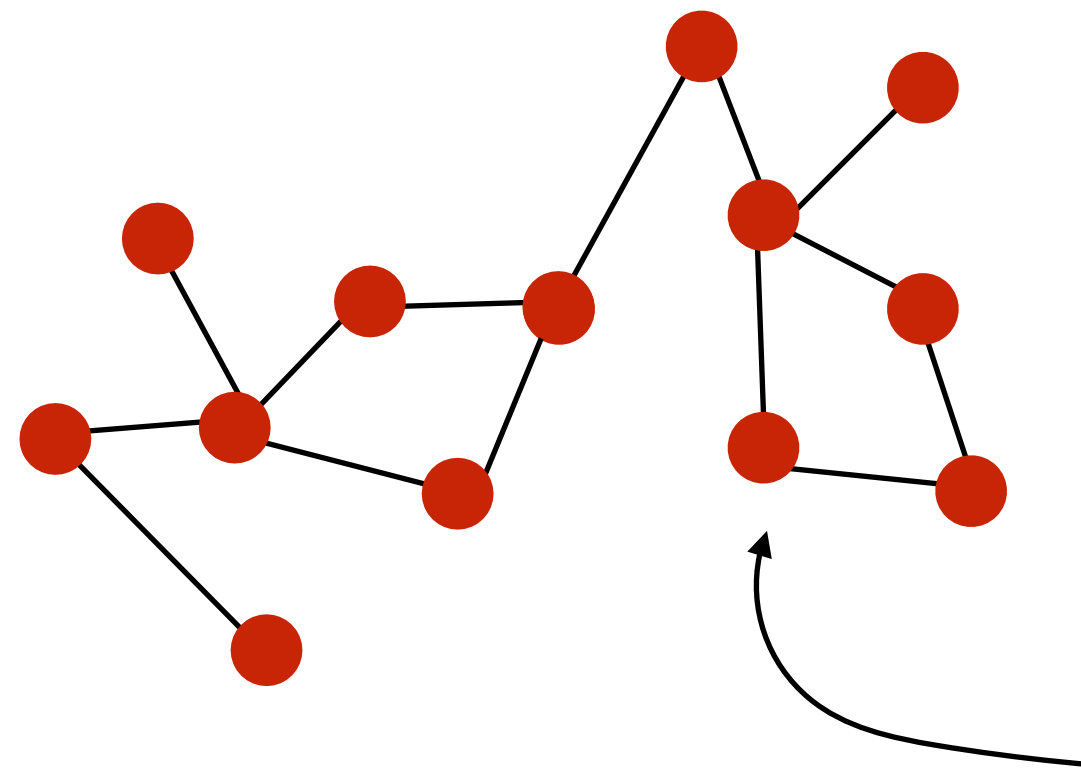


Ribosomes (translation)

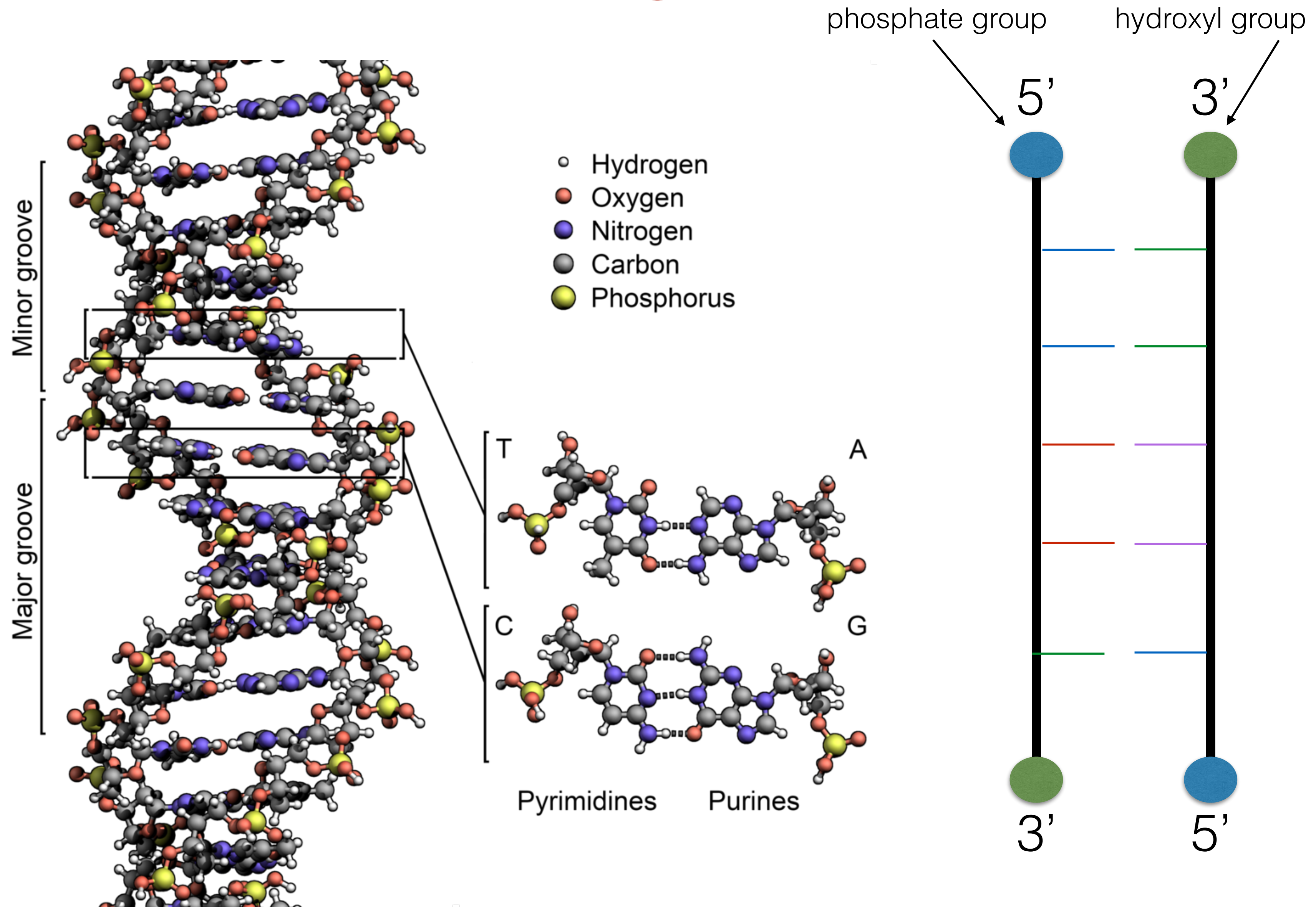
Protein



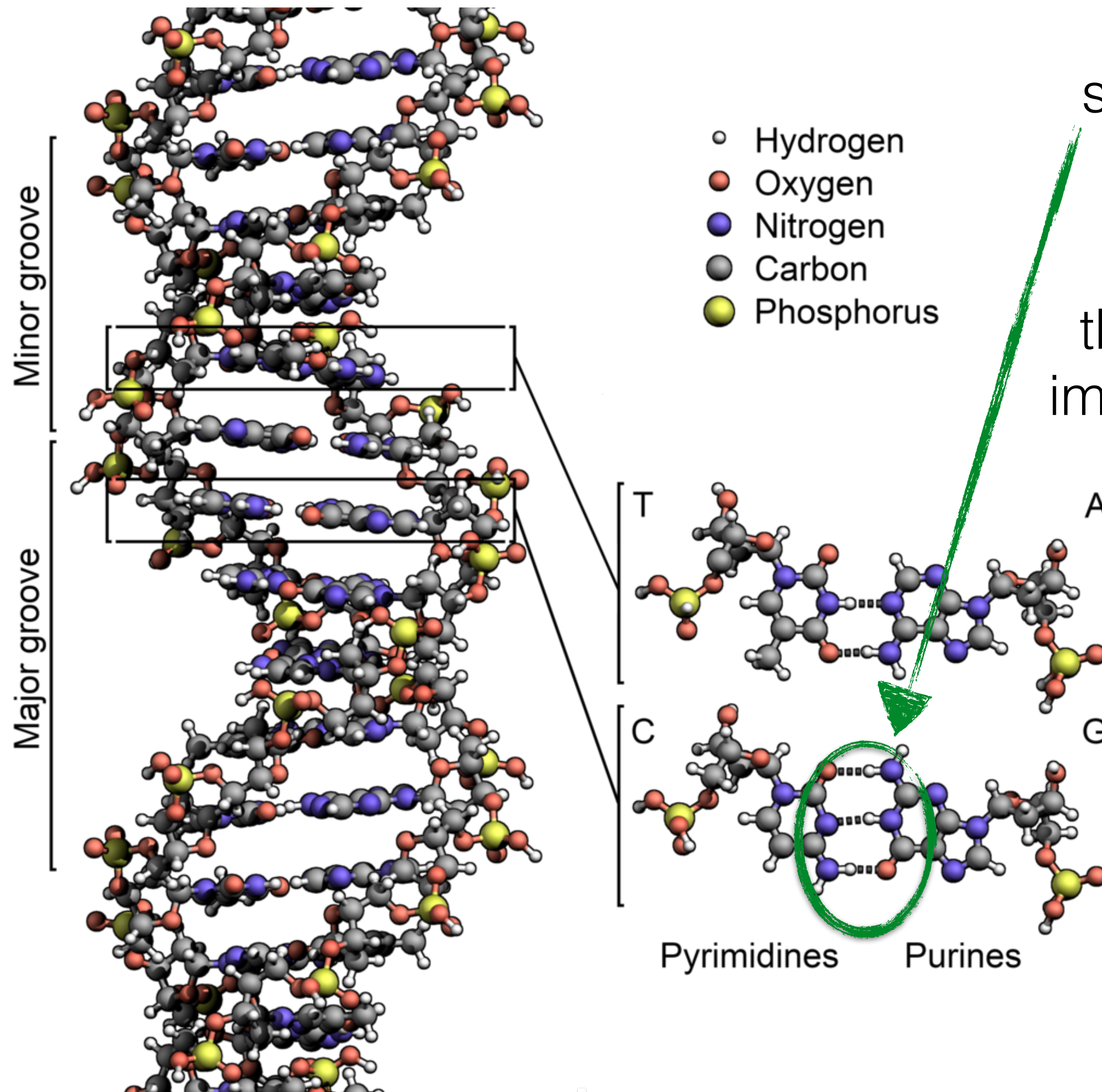
Form networks & pathways; perform a vast set of cellular functions



DNA (the genome)



DNA (the genome)



G-C pairing generally stronger than A-T pairing

Ratio of G+C bases — the “GC content” — is an important sequence feature

DNA (the genome)

gene — a protein-coding gene is transcribed into RNA, which may be translated



“non-coding DNA” — may or may not produce transcripts (e.g. functional non-coding RNA)

In humans, most DNA is “non-coding” ~98%

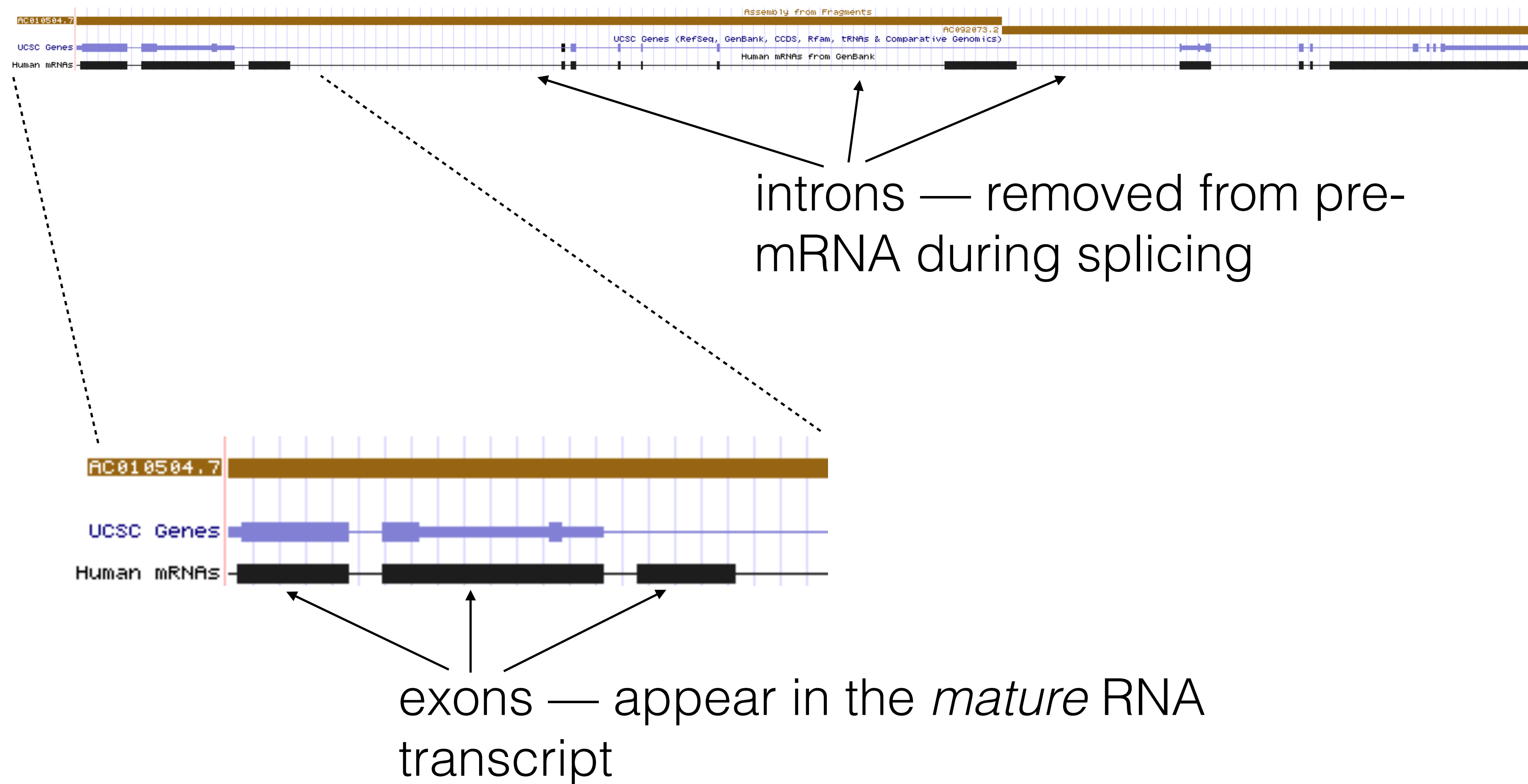
In typical bacterial genome, only small fraction — ~2% — of DNA is “non-coding”

Sometimes referred to as “junk” DNA — much is not, in any way, “junk”

DNA (the genome)

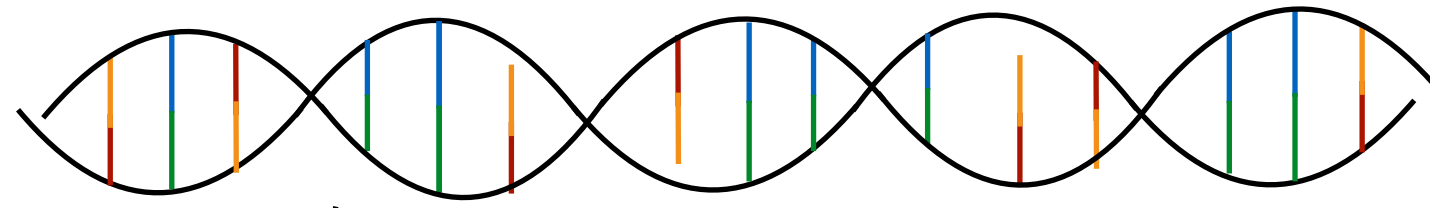
In prokaryotes, genes are typically contiguous DNA segments

In **eukaryotes**, genes can have complex structure



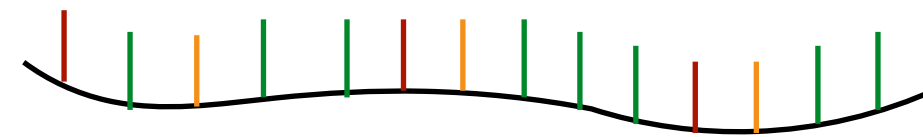
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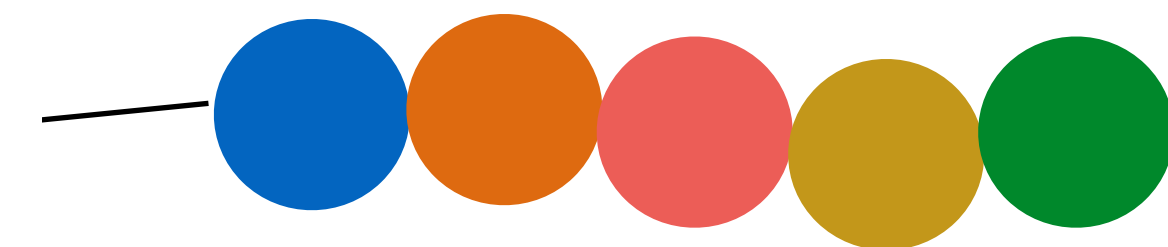
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RNA

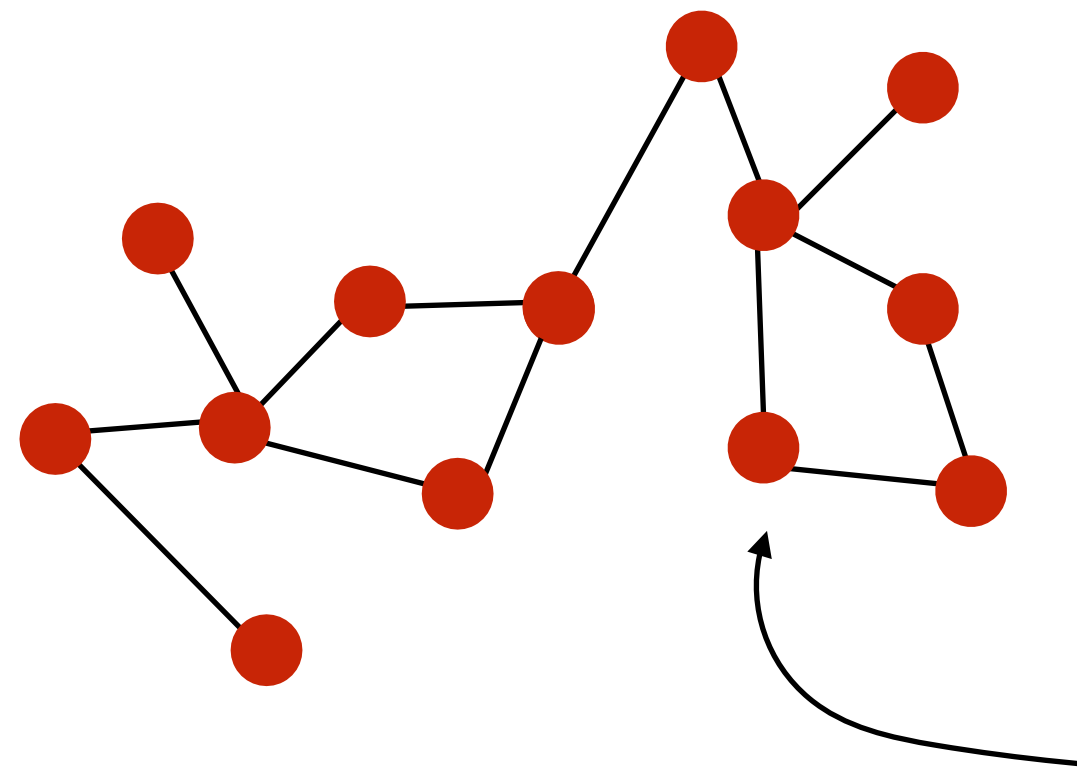


Ribosomes (translation)

Protein

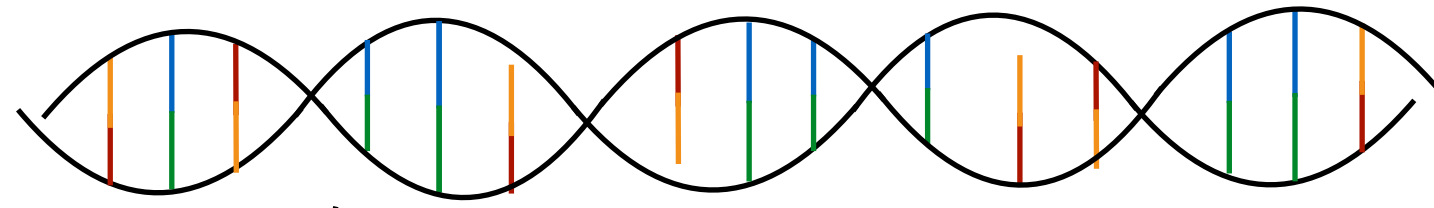


Form networks & pathways; perform a vast set of cellular functions



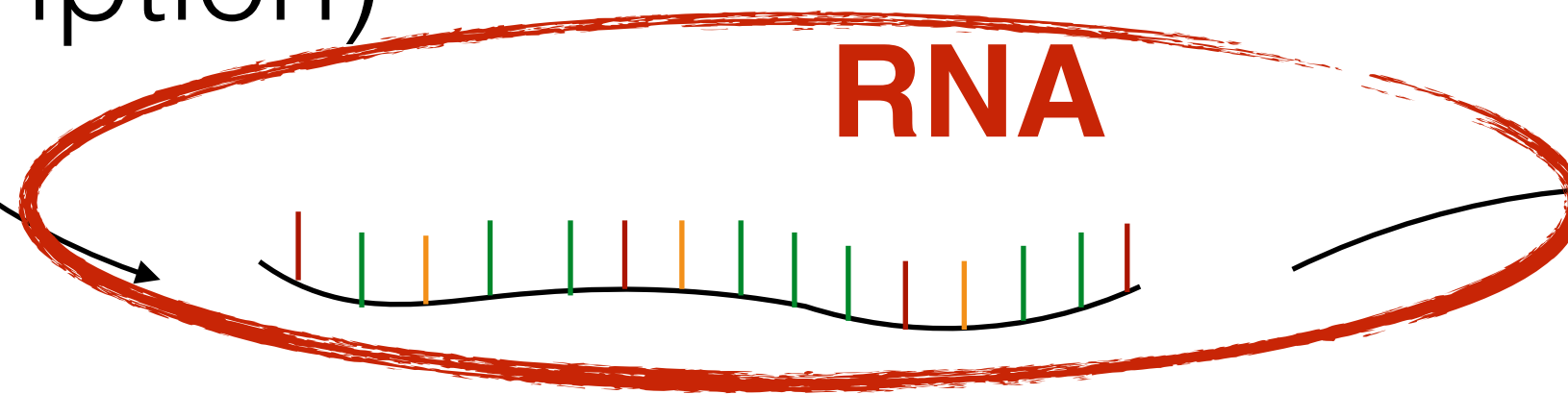
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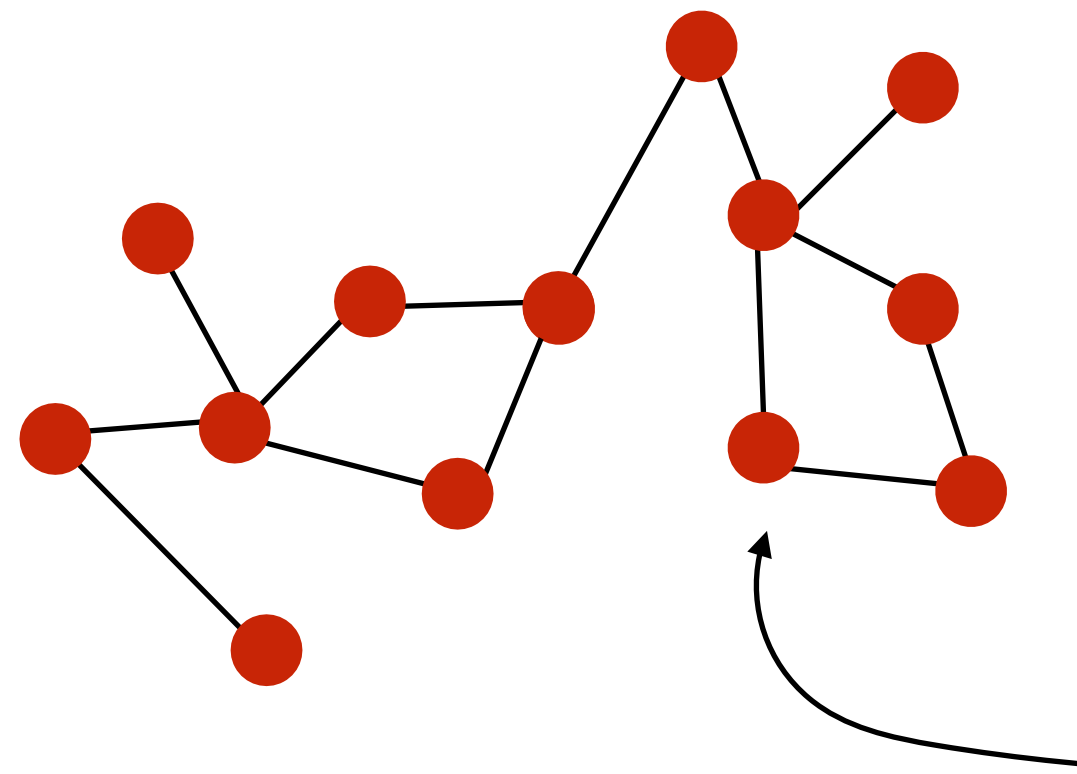
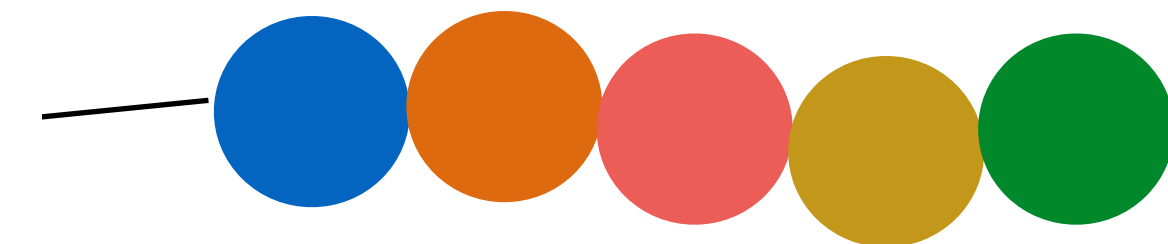
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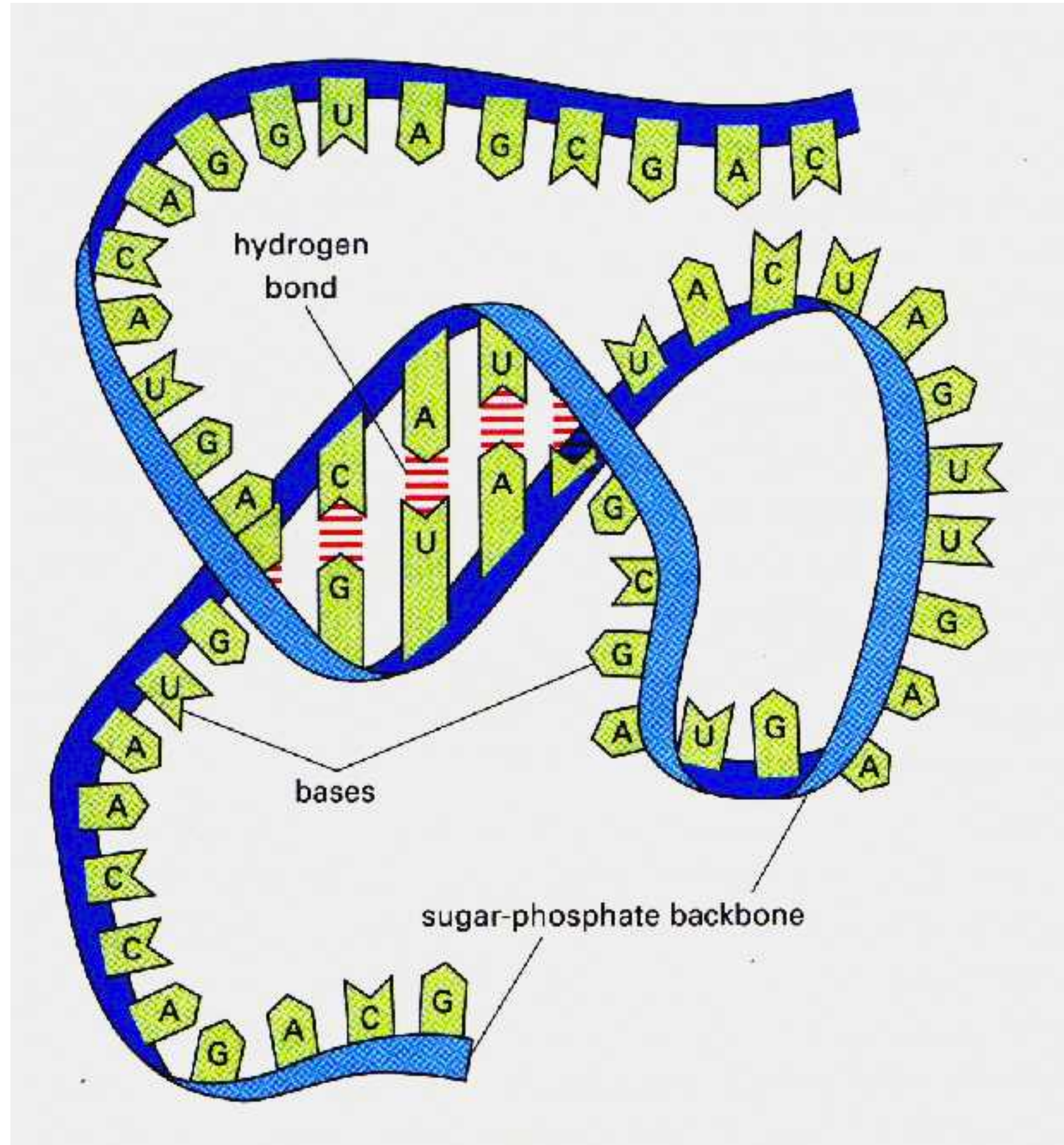
Ribosomes
(translation)

Protein



Form networks &
pathways; perform a
vast set of cellular
functions

RNA



Less regular structure than DNA

Generally a single-stranded molecule

Secondary & tertiary structure can affect function

Act as transcripts for protein, but also perform important functions themselves

Same “alphabet” as DNA, except thymine replaced by uracil

RNA Splicing

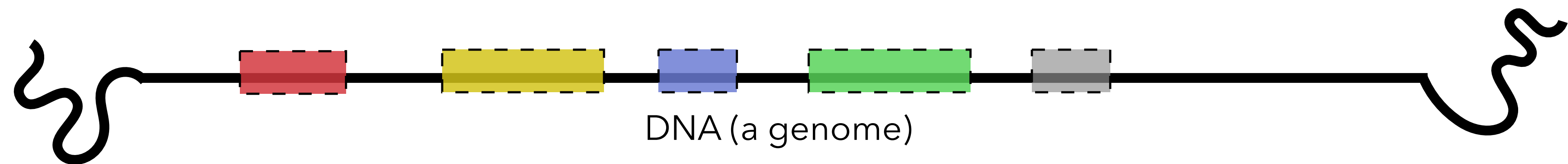
DNA transcribed into pre-mRNA

Some “processing” occurs **capping** & **polyadenylation**

Introns removed from pre-mRNA resulting in mature mRNA

mature mRNA

pre-mRNA



RNA Splicing

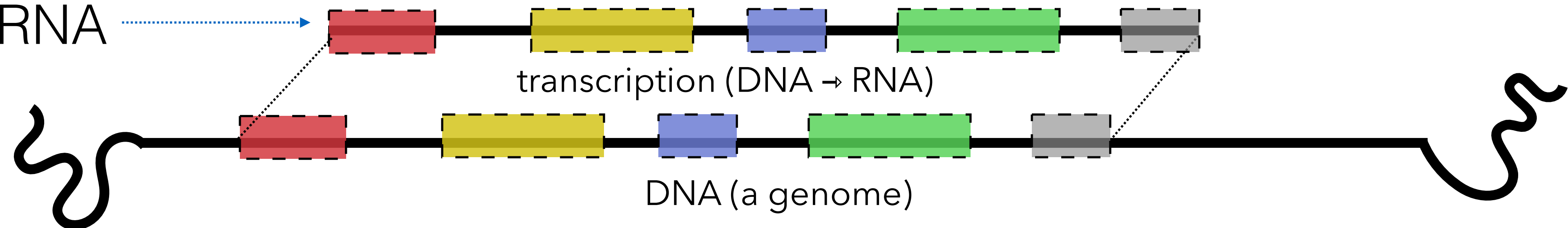
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mature mRNA

pre-mRNA

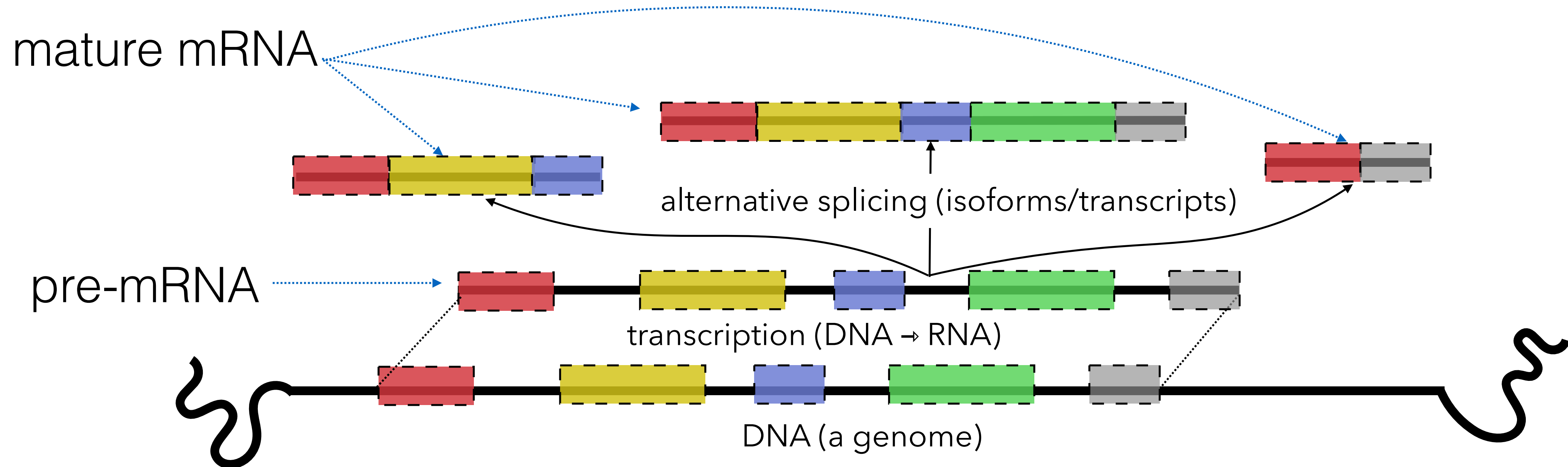


RNA Splicing

DNA transcribed into pre-mRNA

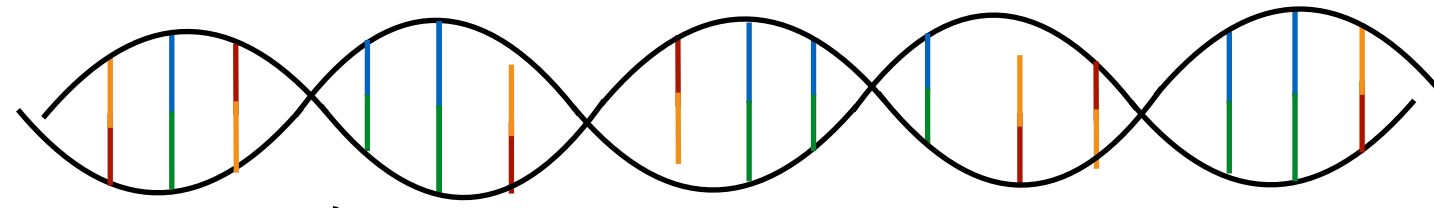
Some “processing” occurs **capping** & **polyadenylation**

Introns removed from pre-mRNA resulting in mature mRNA



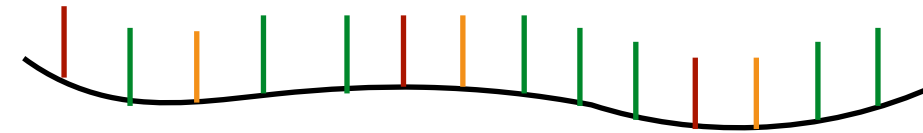
“Flow” of information in the cell

DNA



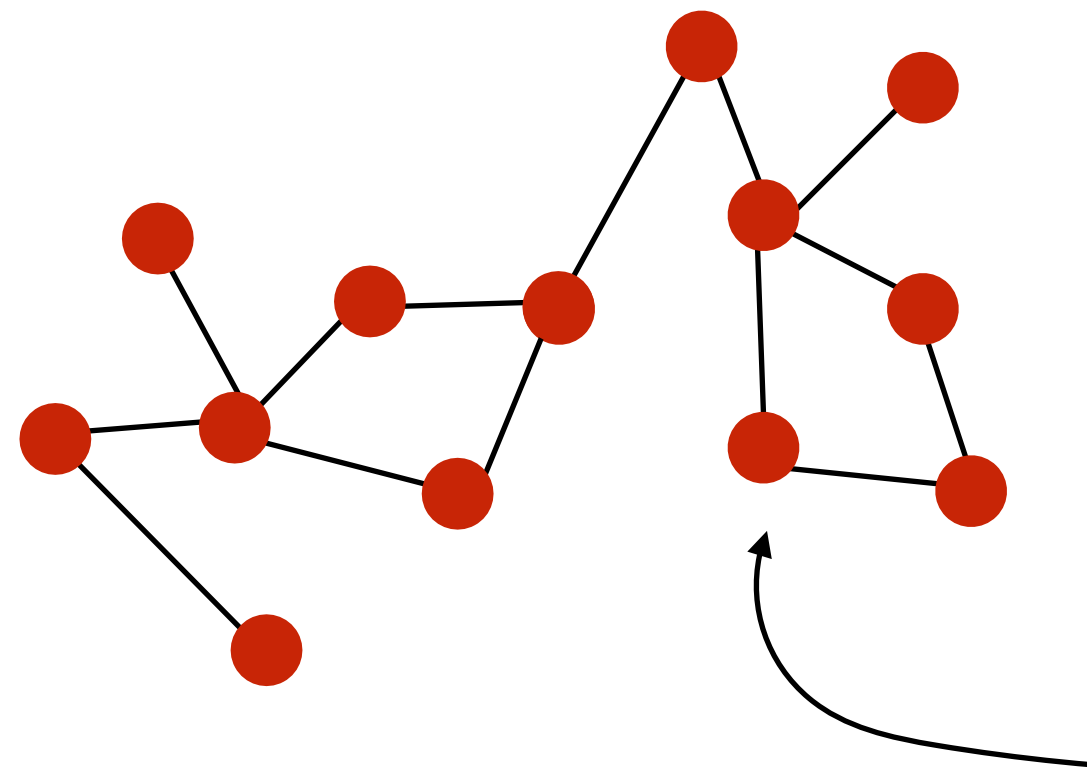
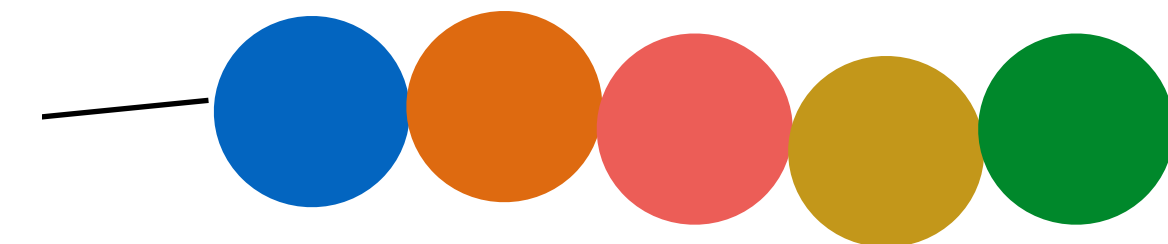
RNA Polymerase
(transcription)

RNA



Ribosomes
(translation)

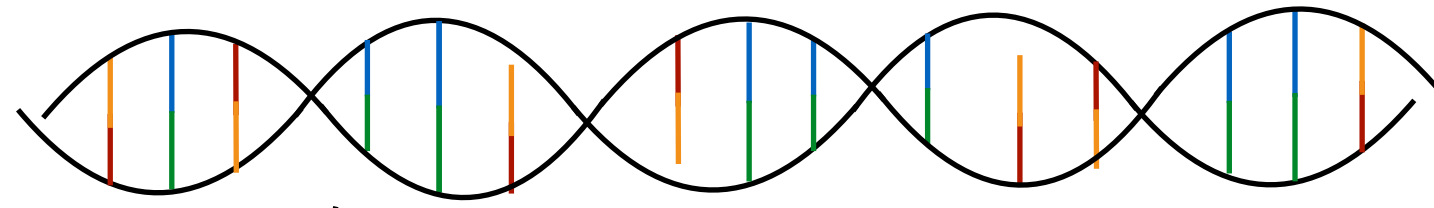
Protein



Form networks &
pathways; perform a
vast set of cellular
functions

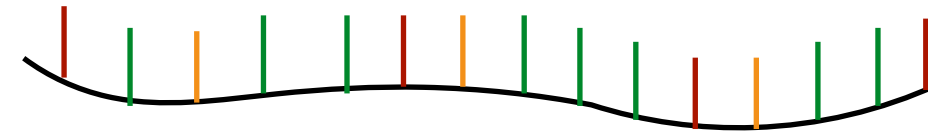
“Flow” of information in the cell

DNA



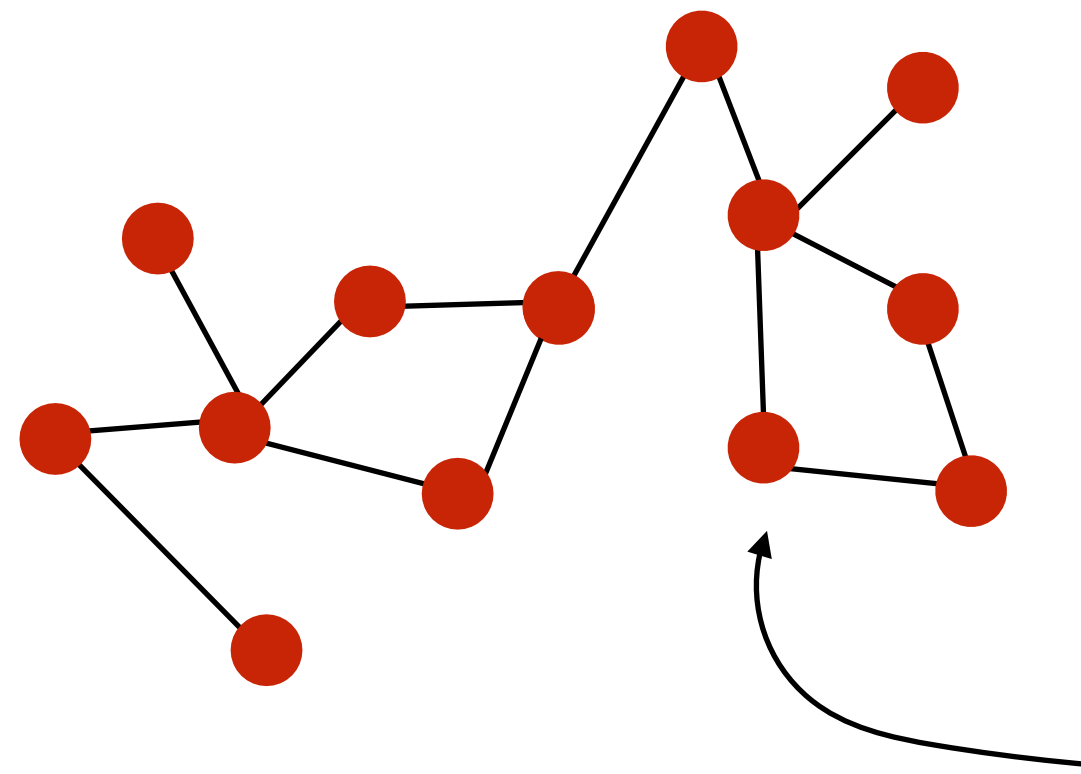
RNA Polymerase
(transcription)

RNA



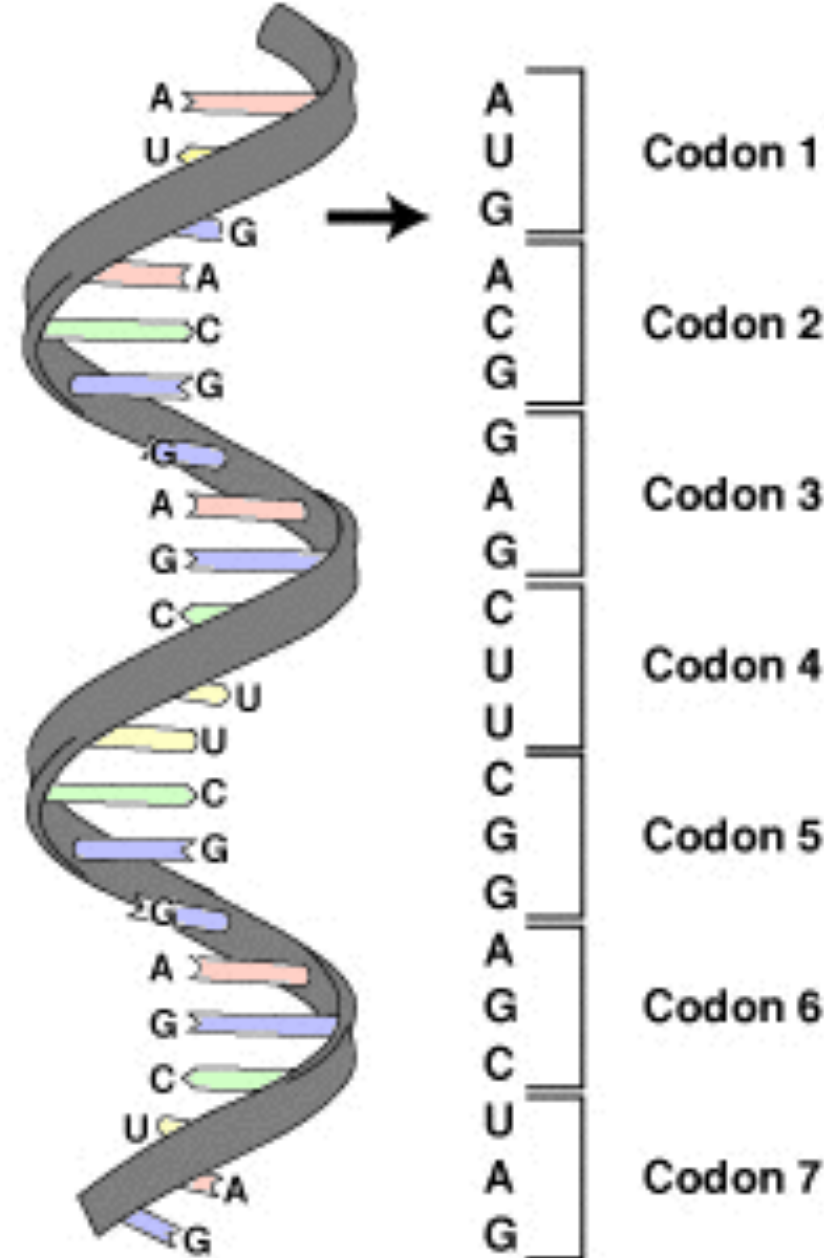
Ribosomes
(translation)

Protein



Form networks &
pathways; perform a
vast set of cellular
functions

Protein



Triplets of mRNA bases (codons) correspond to specific amino acids

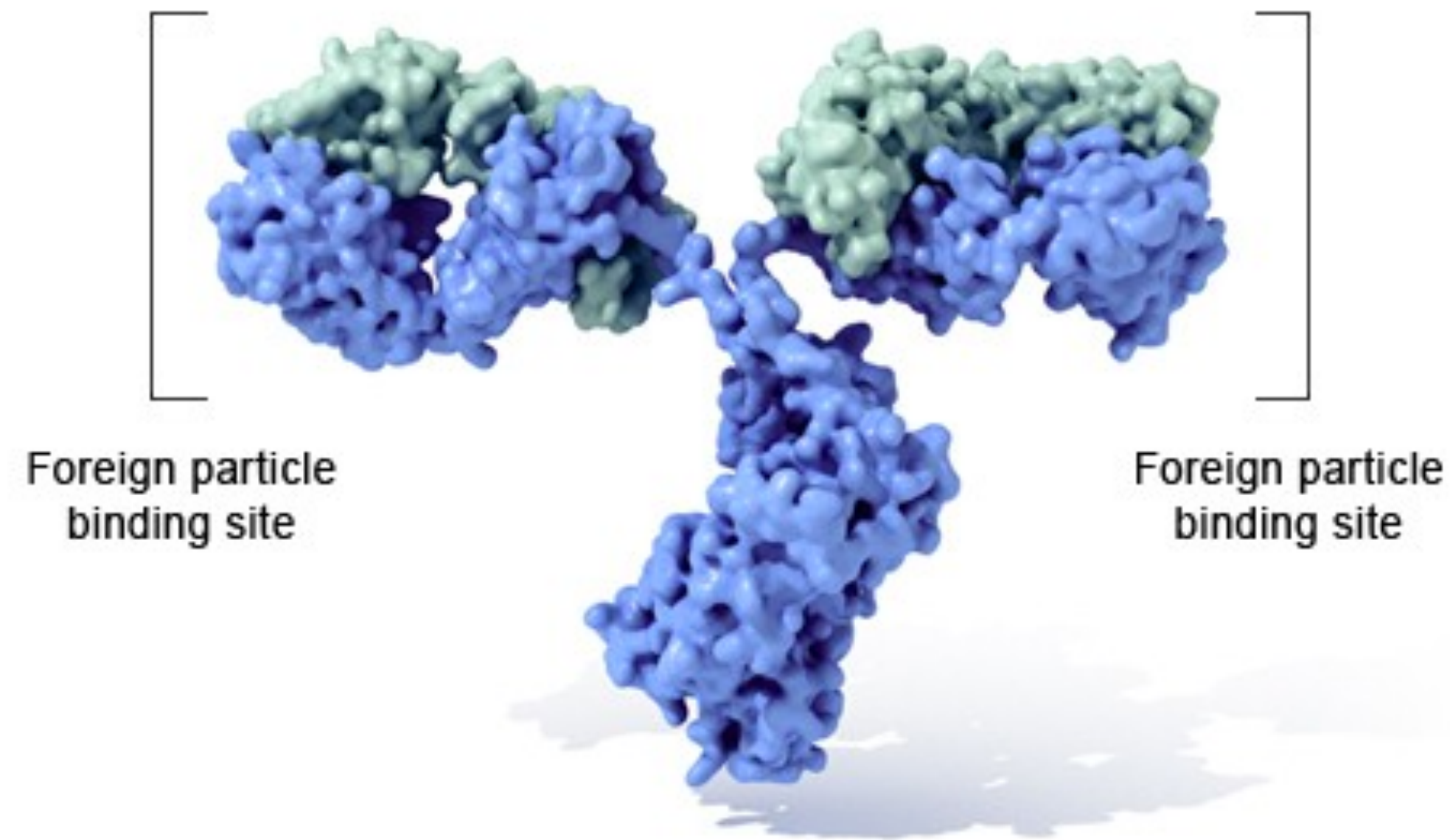
This mapping is known as the “genetic code” — an *almost* law of molecular Biology

Inverse table (compressed using **IUPAC notation**)

Amino acid	Codons	Compressed	Amino acid	Codons	Compressed
Ala/A	GCU, GCC, GCA, GCG	GCN	Leu/L	UUA, UUG, CUU, CUC, CUA, CUG	YUR, CUN
Arg/R	CGU, CGC, CGA, CGG, AGA, AGG	CGN, MGR	Lys/K	AAA, AAG	AAR
Asn/N	AAU, AAC	AAY	Met/M	AUG	
Asp/D	GAU, GAC	GAY	Phe/F	UUU, UUC	UUY
Cys/C	UGU, UGC	UGY	Pro/P	CCU, CCC, CCA, CCG	CCN
Gln/Q	CAA, CAG	CAR	Ser/S	UCU, UCC, UCA, UCG, AGU, AGC	UCN, AGY
Glu/E	GAA, GAG	GAR	Thr/T	ACU, ACC, ACA, ACG	ACN
Gly/G	GGU, GGC, GGA, GGG	GGN	Trp/W	UGG	
His/H	CAU, CAC	CAY	Tyr/Y	UAU, UAC	UAY
Ile/I	AUU, AUC, AUA	AUH	Val/V	GUU, GUC, GUA, GUG	GUN
START	AUG		STOP	UAA, UGA, UAG	UAR, URA

Protein

Immunoglobulin G (IgG)



Perform vast majority of intra & extra cellular functions

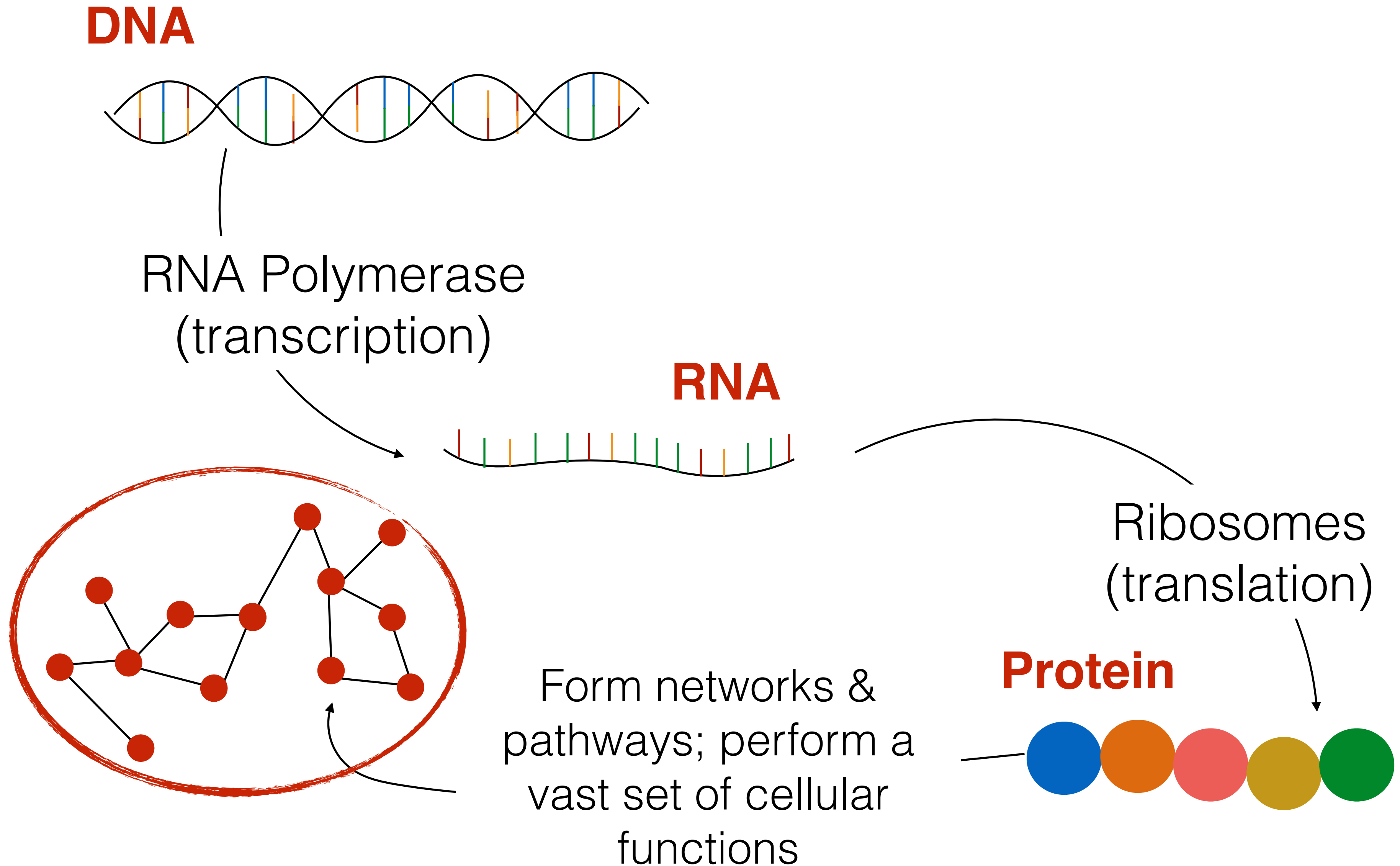
Can range from a few amino acids to *very* large and complex molecules

Can bind with other proteins to form protein complexes

U.S. National Library of Medicine

The shape or *conformation* of a protein is intimately tied to its function. Protein shape, therefore, is strongly conserved through evolution — even more so than sequence. A protein can undergo sequence mutations, but fold into the same or a similar shape and still perform the same function.

“Flow” of information in the cell



An extension of this DNA → RNA → protein cartoon ... retroviruses

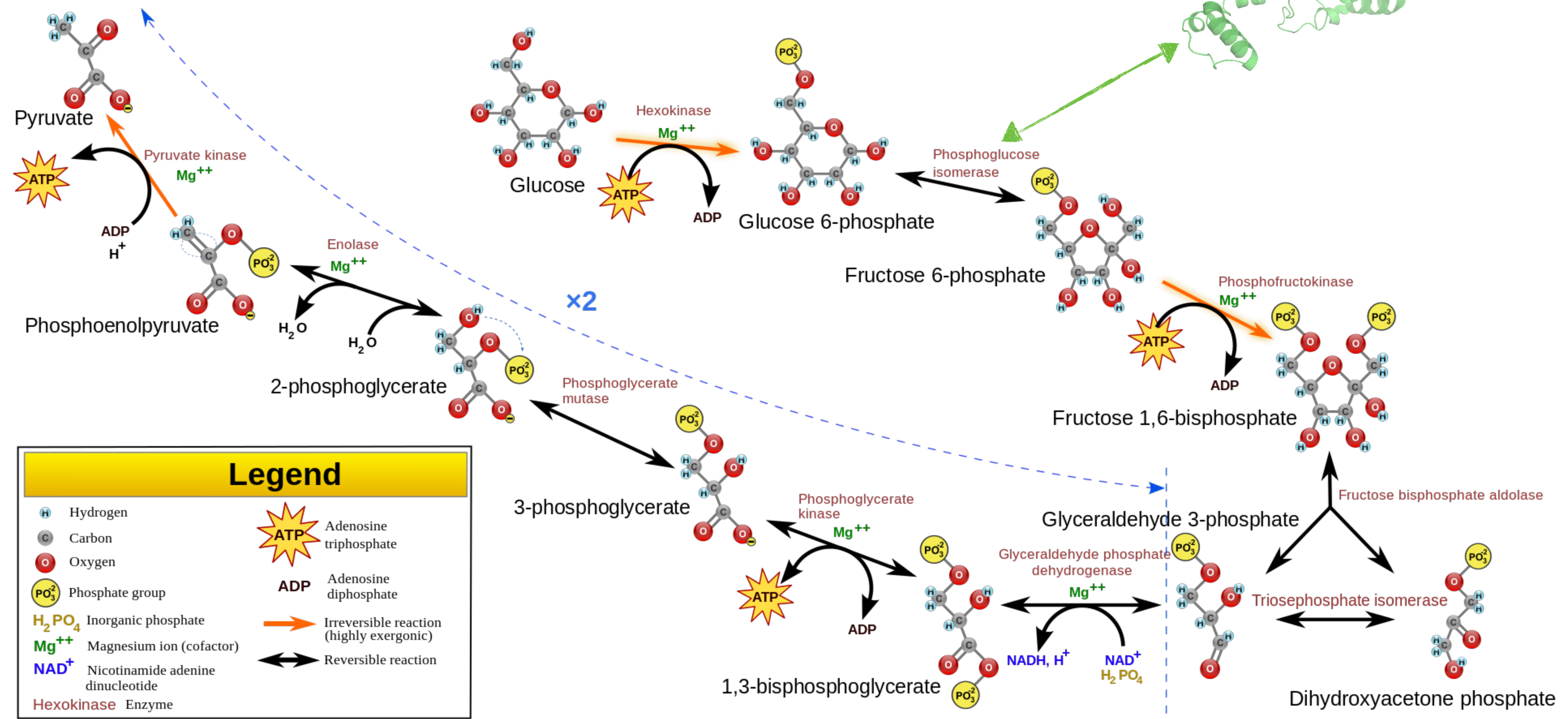
Glycolysis Pathway

Converts glucose → pyruvate

Generates ATP (“energy currency” of the cell)

phosphoglucose isomerase

this is an **example**, no need to memorize this Bio.



Some Interesting Facts

Organism	Genome size	# of genes (protein coding)
ϕ X174 (<i>E. coli</i> virus)	~5kb	11
<i>E. coli</i> K-12	~4.6Mb	~4,300
Fruit Fly	~122Mb	~17,000
Human	~3.05Gb	~21,000
Mouse	~2.8Gb	~23,000
<i>P. abies</i> (a spruce tree)	~19.6Gb	~28,000

No strong link between genome size & phenotypic complexity

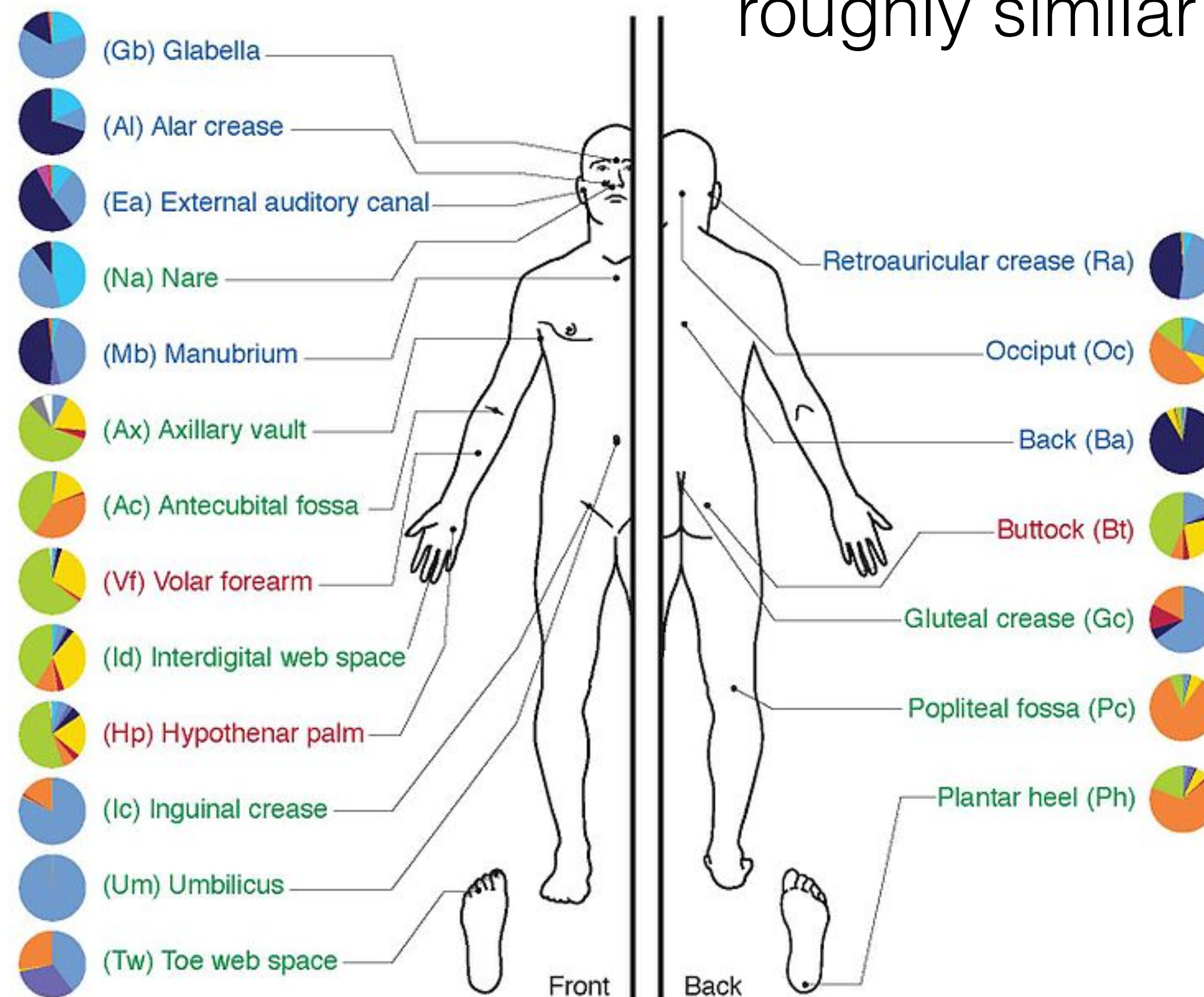
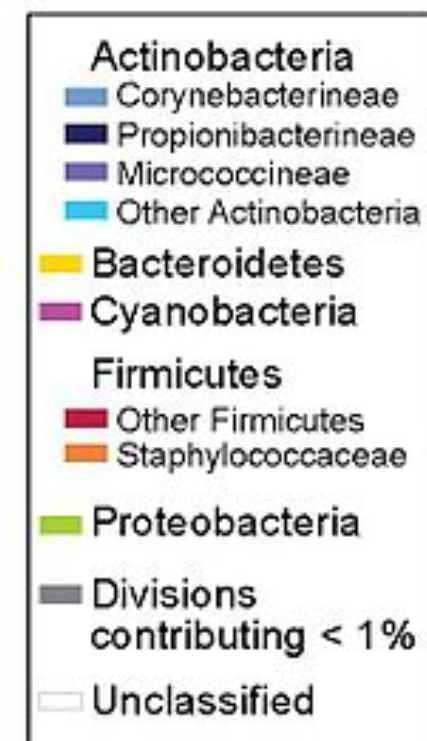
Plants can have **huge** genomes

Some Interesting Facts

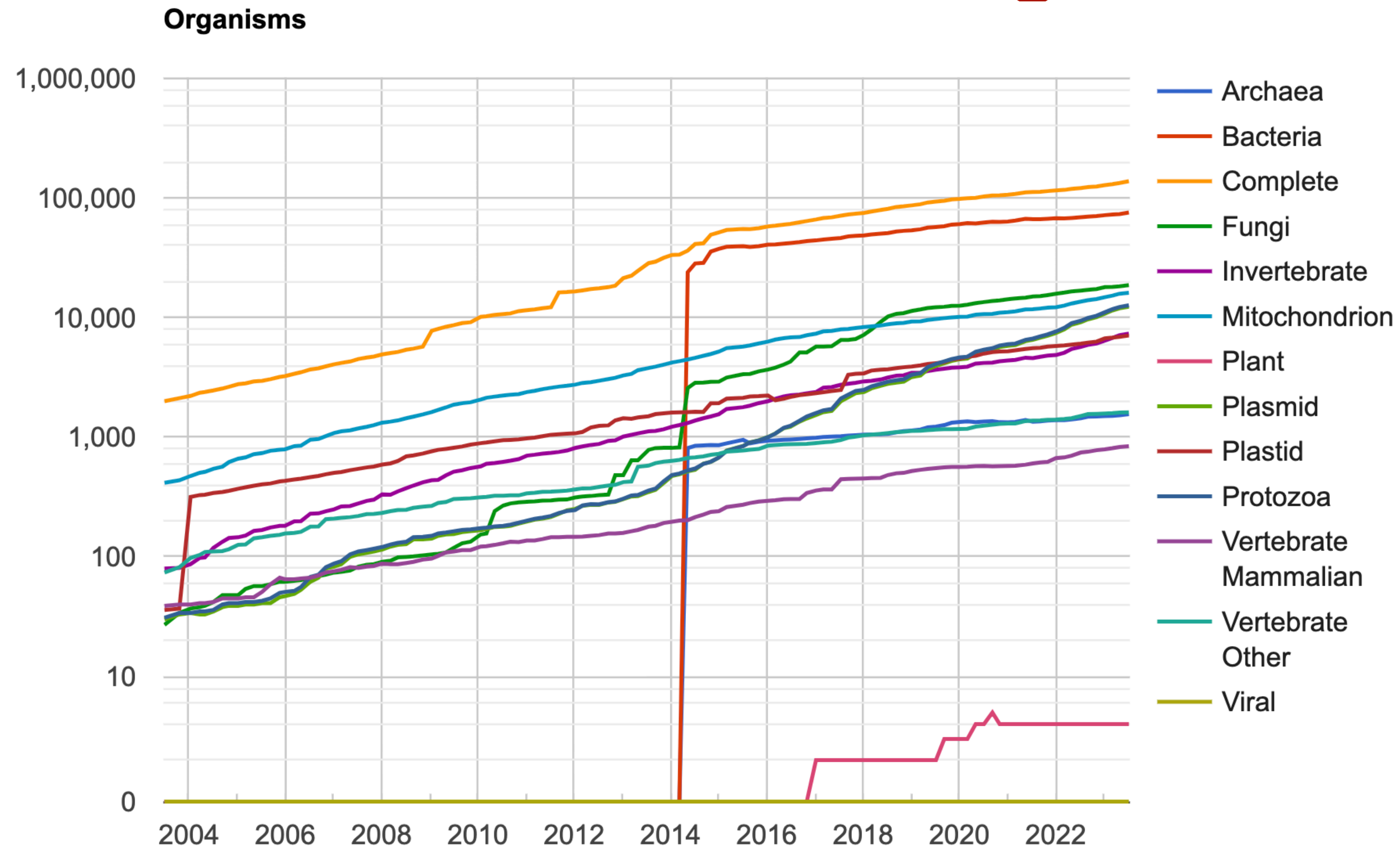
Your body is home to a vast microbial community

Bacterial cells and human cells are present in roughly similar numbers

This community of organisms is called the microbiota



Some Interesting Facts



. . . Out of 8.7 ± 1.3 Mil*

Vast majority of species unsequenced & *can not be cultivated in a lab* (one of the many motivations for metagenomics)

*Mora, Camilo, et al. "How many species are there on Earth and in the ocean?." PLoS biology 9.8 (2011): e1001127.

A lens for every topic

For every genomics problem, ask:

QUESTION

What biological claim
do we want to make?



DATA

What does the
technology
reveal—and what does
it miss?



METHOD

Which abstraction turns
observations into an
answer?

Next: sequence data as computational objects.